



Qudelix-5K DAC/AMP

User's Guide



JULY 2026

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Key Specification

- **Bluetooth 5.0**
 - Class 2 (2.5 mW / 4 dBm) up to 10m
 - 2.4GHz
- **Qualcomm QCC5124 Bluetooth Chipset**
- **Dual ES9219 SABRE HiFi® DAC**
 - 3.5mm output works with single DAC only.
 - Dual DAC is available with 2.5mm output.
- **High Precision MEMS Audio Clock**
- **aptX Adaptive, LDAC, AAC, aptX-HD, aptX, SBC**
- **High-Resolution USB DAC 96 kHz / 24-bit**
 - USB Audio Class 1.0
 - Windows/macOS/Linux (No Device Driver)
 - Android Device through C-to-C or OTG cable
- **20-band Double Precision GEQ/PEQ**
- **10-band Left/Right Independent GEQ/PEQ**
- **High Sensitivity 3-D LDS (Laser Direct Structuring) Antenna**
 - Bluetooth Range more than 10 meters
- **Qualcomm® cVc™ noise cancellation technology**
- **High Sensitivity MEMS Microphone**
- **Four Multi-functional Buttons & Two LEDs**
- **Qudelix iOS/Android mobile app**
- **Qudelix Chrome app for PC/macOS**
- **Over-the-Air firmware update**
- **Material**
 - Plastic Body (Black) with anti-scratch UV Coating
 - Aluminium Clip (Dark Gray)
- **Size 52.8 x 26.7 x 15.6mm (including Clip)**
- **Weight 25 g**

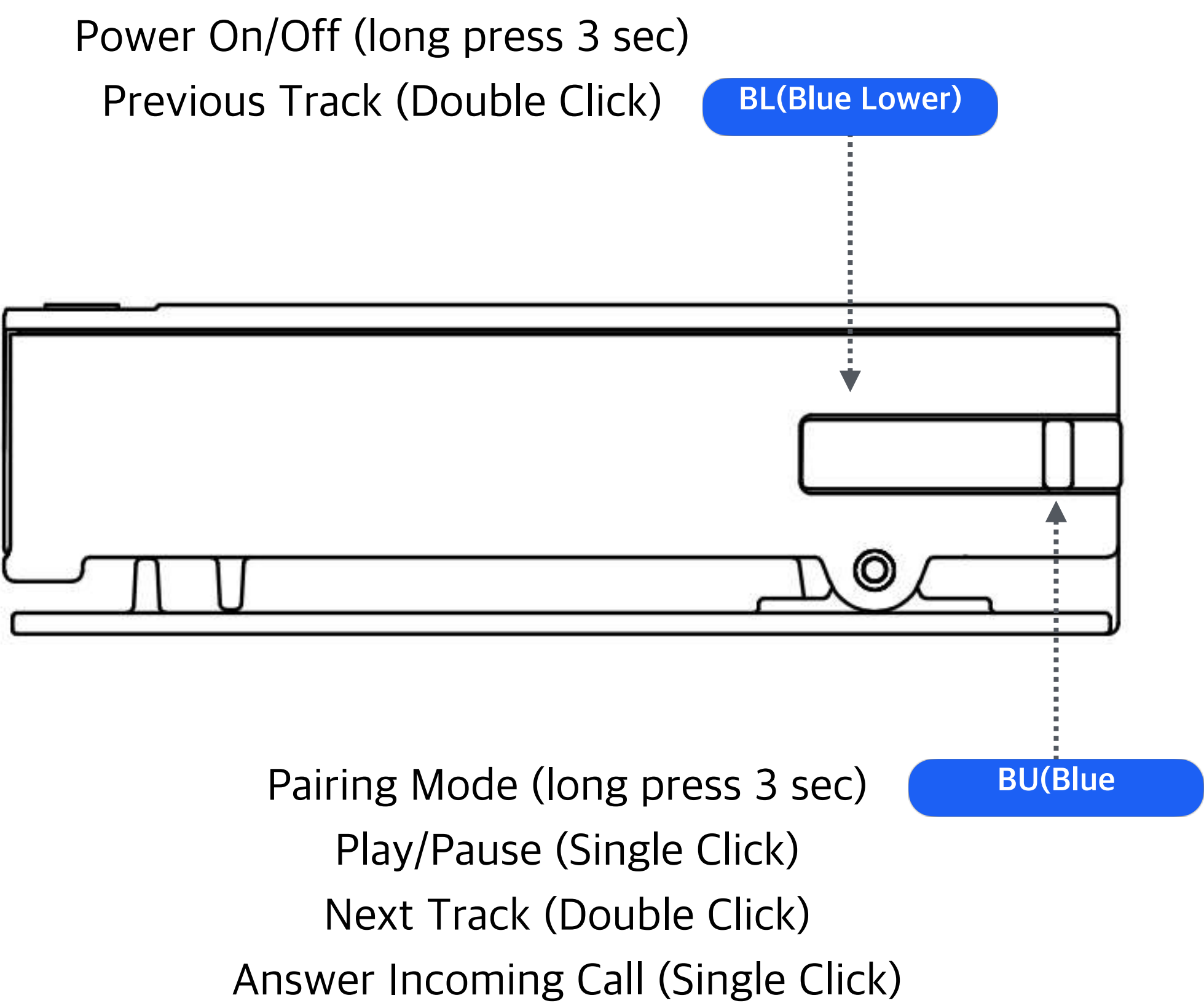


Key Specification

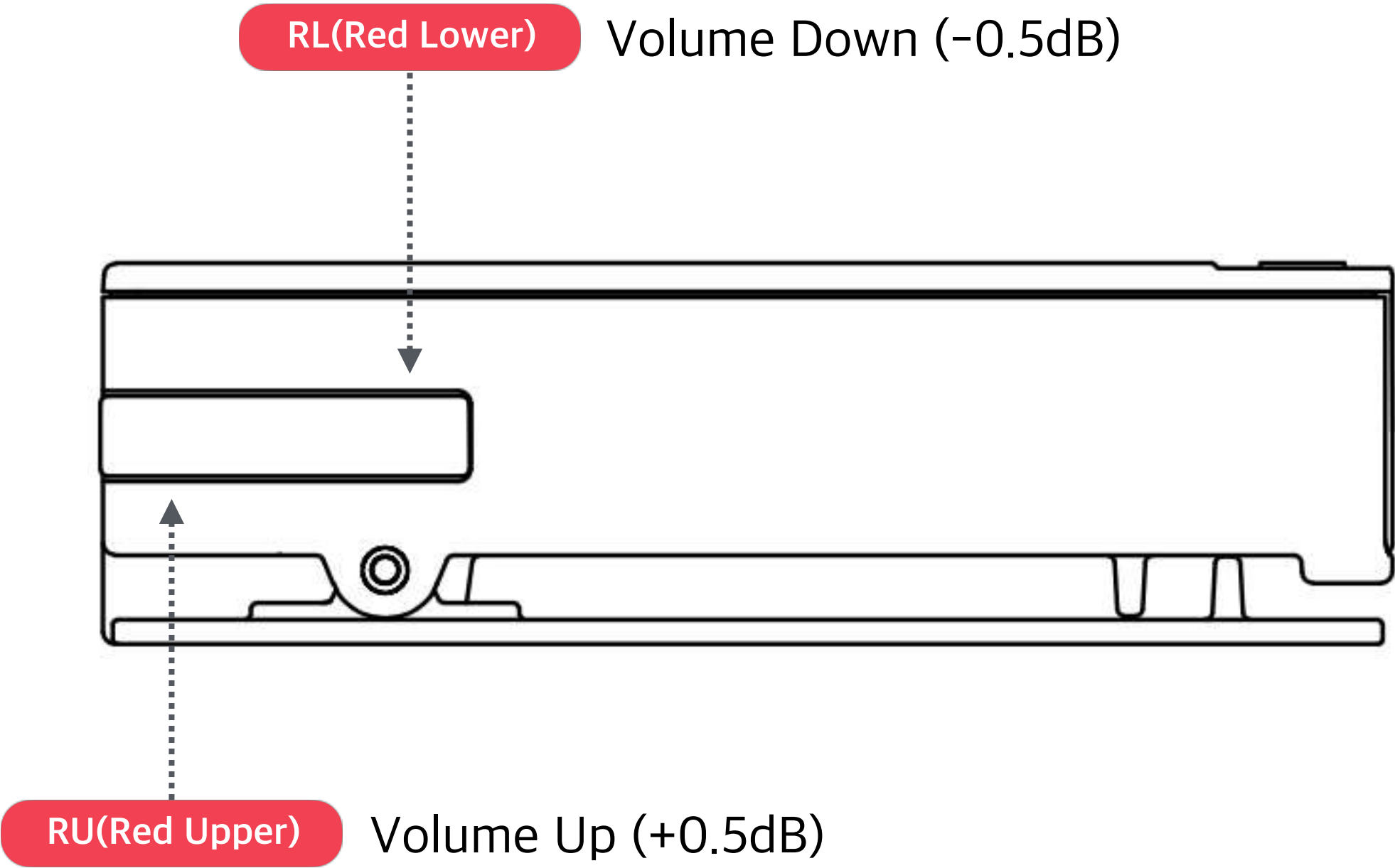
- **Output Power**
 - 3.5mm 80mW per channel
 - 2.5mm 240mW per channel
- **SNR (A-weighted)**
 - 3.5mm -118dB
 - 2.5mm -122dB
- **THD+N**
 - 3.5mm 0.004%
 - 2.5mm 0.002%
- **Separation**
 - 3.5mm 79dB
 - 2.5mm 117dB (1kHz/32-ohm)
- **Output Impedance**
 - Less than 1-ohm



Buttons (with BLUE LED)

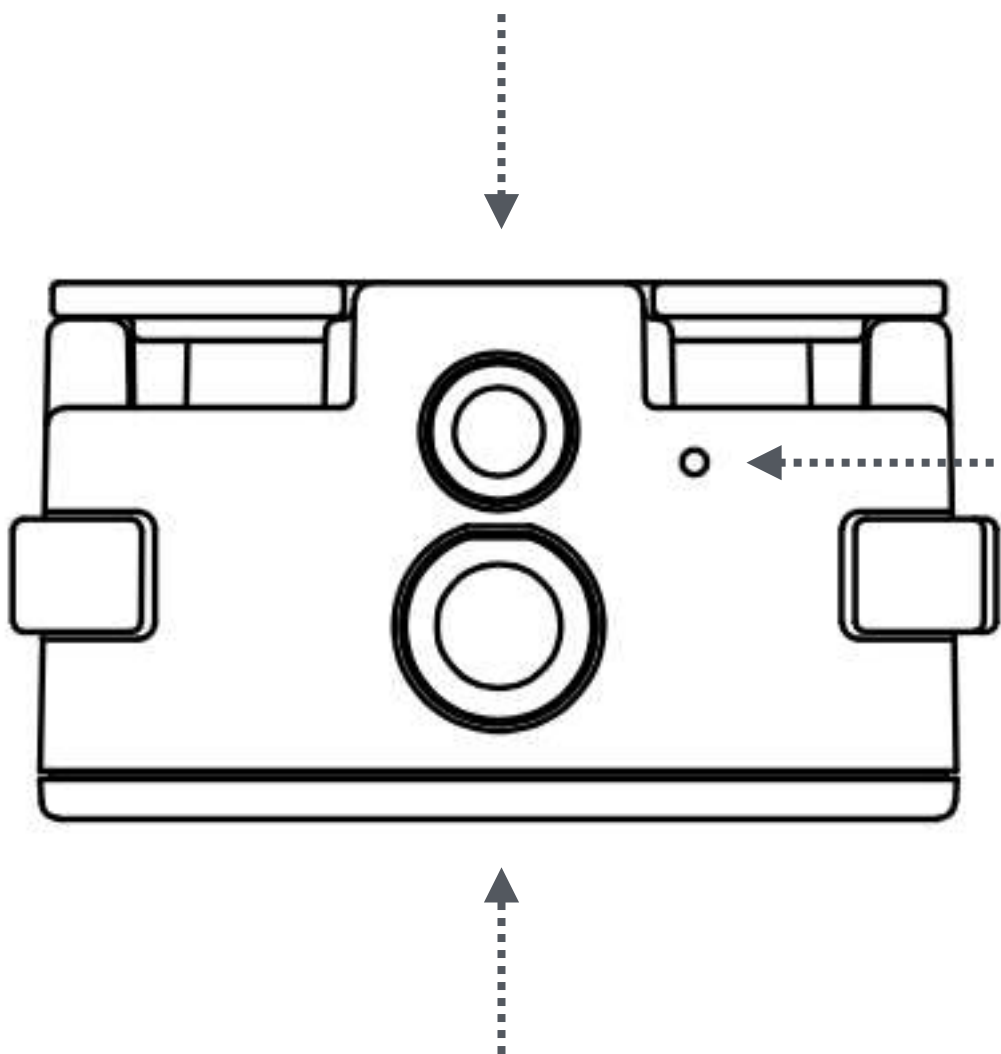


Buttons (with RED LED)



Audio Output & Microphone

2.5mm Balanced Audio Output
Max. 4 V RMS (240 mW/Ch)



Built-In Microphone

3.5mm Unbalanced Audio Output
Max. 2 V RMS (80 mW/Ch)



Pairing with Android

1. Power On the 5K.

- Press **BL(Blue Lower)** button for more than 3 sec.

2. Enter Pairing Mode

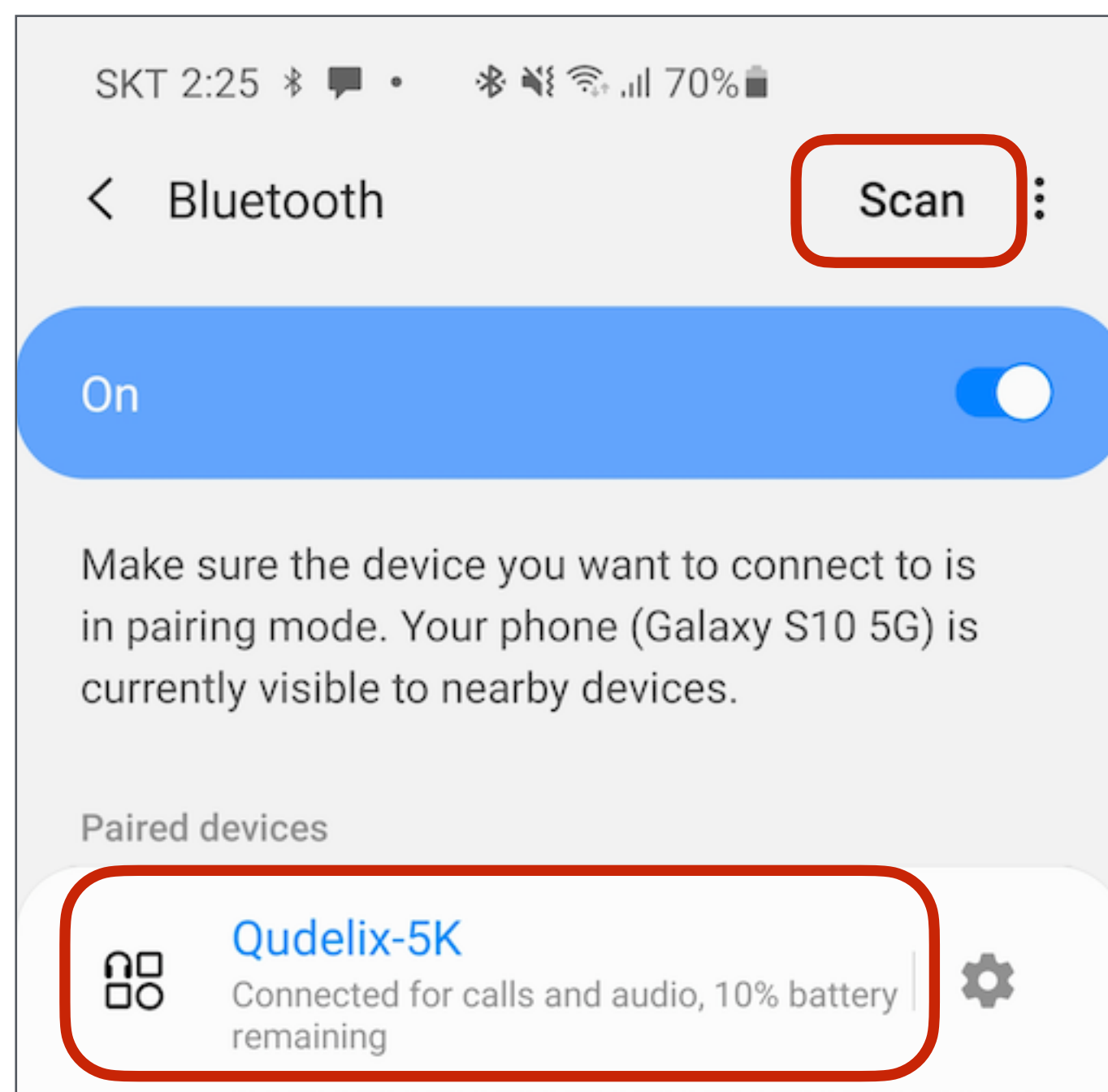
- Press **BU(Blue Upper)** button for more than 3 sec.
- **Red, Blue** LED blink when pairing mode.

3. Go to the Android Bluetooth menu

- Scan Bluetooth devices.
- Select Qudelix-5K in the Available devices list.
- If Qudelix-5K does not appear, please toggle the Bluetooth button OFF and ON, then try scanning again.

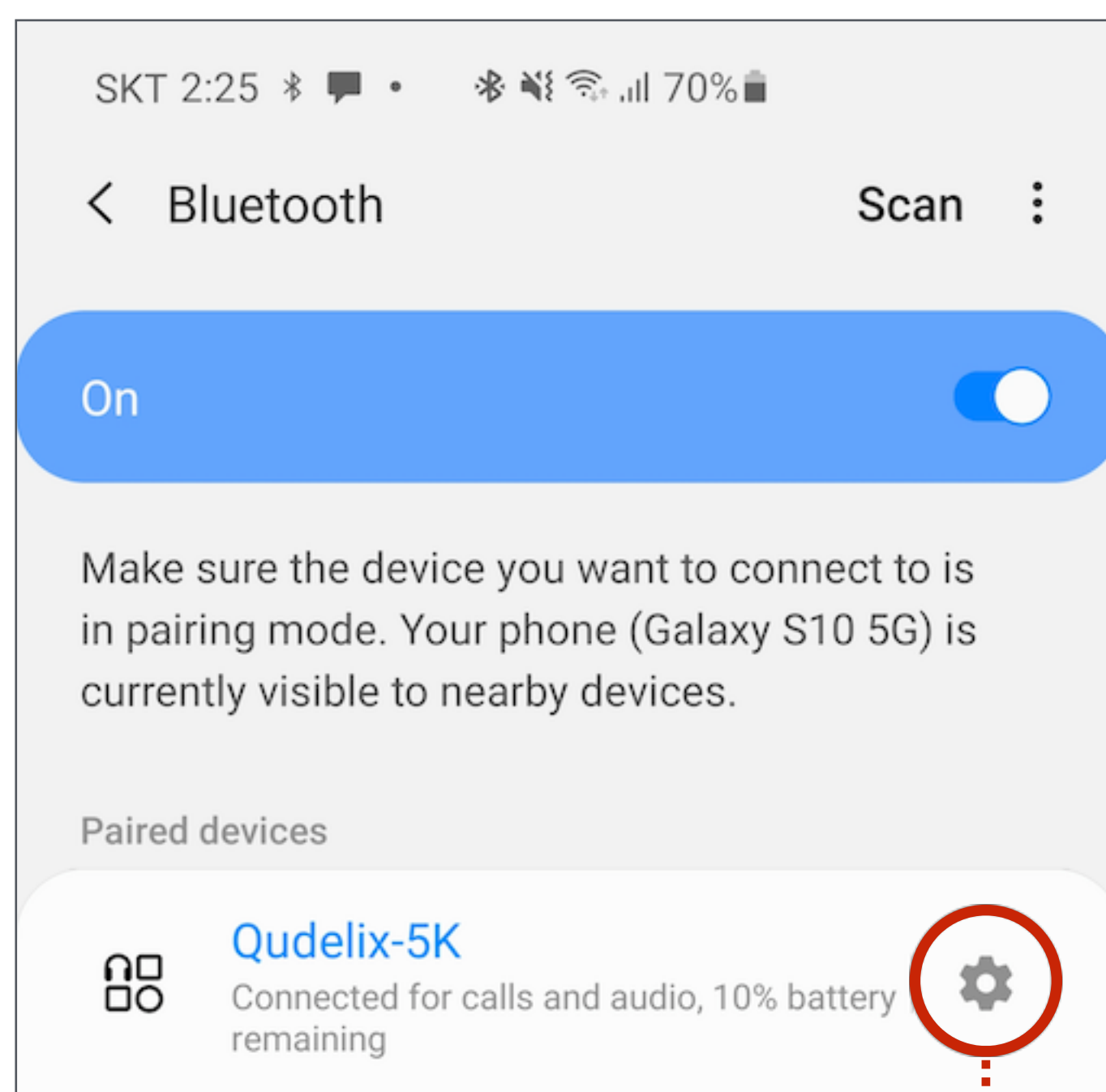
4. Paired

- Once paired successfully, you will see Qudelix-5K in the Paired devices list.

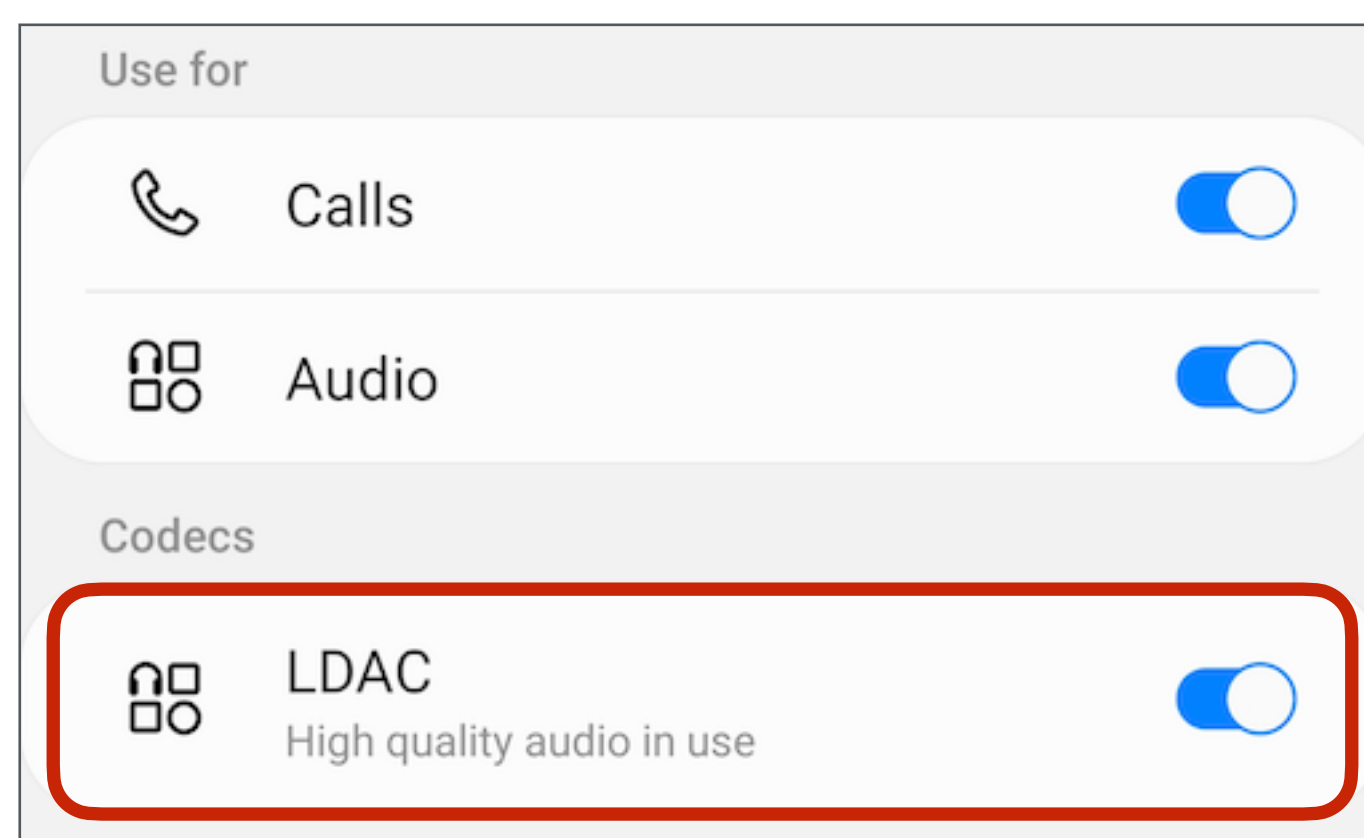


Android - LDAC

- Android 8.0 or higher supports **SONY LDAC**, a high-quality audio codec.
- After pairing the device with an Android smartphone, please enable LDAC in the Android Bluetooth menu.
 - Go to the paired devices list
 - Select the option button for Qudelix-5K
 - Enable LDAC for high-quality audio streaming



Option Button



***Enable Sony LDAC after pairing**

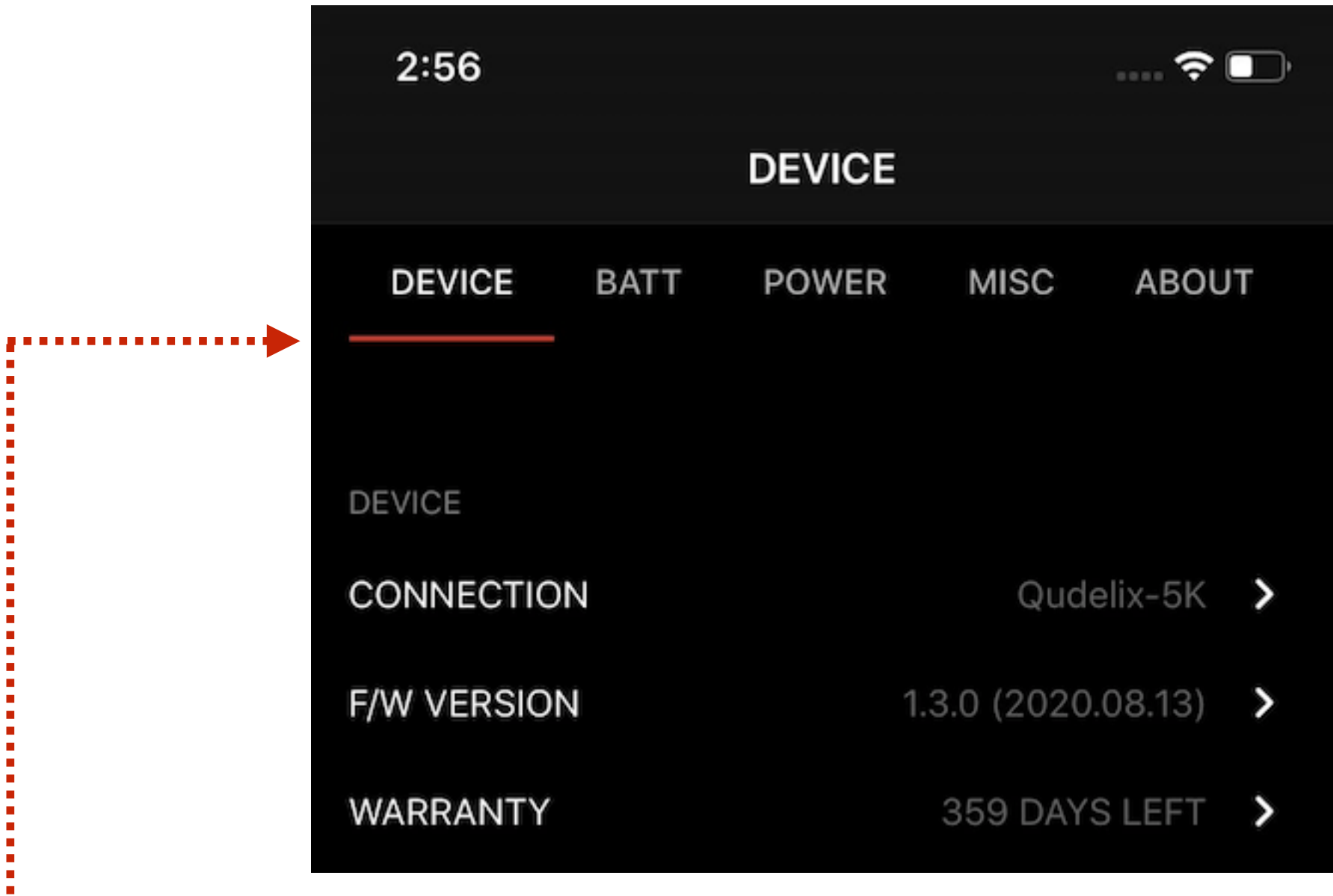
Mobile App - Android



- Install Android Mobile App (requires Android 8.0 or higher)
- https://play.google.com/store/apps/details?id=com.qudelix.qudelix&hl=en_US



- The Qudelix App has nothing to do with Bluetooth Audio Streaming. You can get audio streaming over Bluetooth without Qudelix App. However, we strongly recommend installing and using the app to fully experience the feature-rich nature of Qudelix-5K.
- The 5K supports an app connection with only one device at a time.
- Android App uses the BR/EDR link. So, Android requires the 5K device paired to the system before making the app connection.



*The Red Bar indicates the app connected to the device; gray otherwise.

Pairing with iOS

1. Power On the 5K.

- Press **BL(Blue Lower)** button for more than 3 sec.

2. Enter Pairing Mode

- Press **BU(Blue Upper)** button for more than 3 sec.
- **Red, Blue** LED blink when pairing mode.

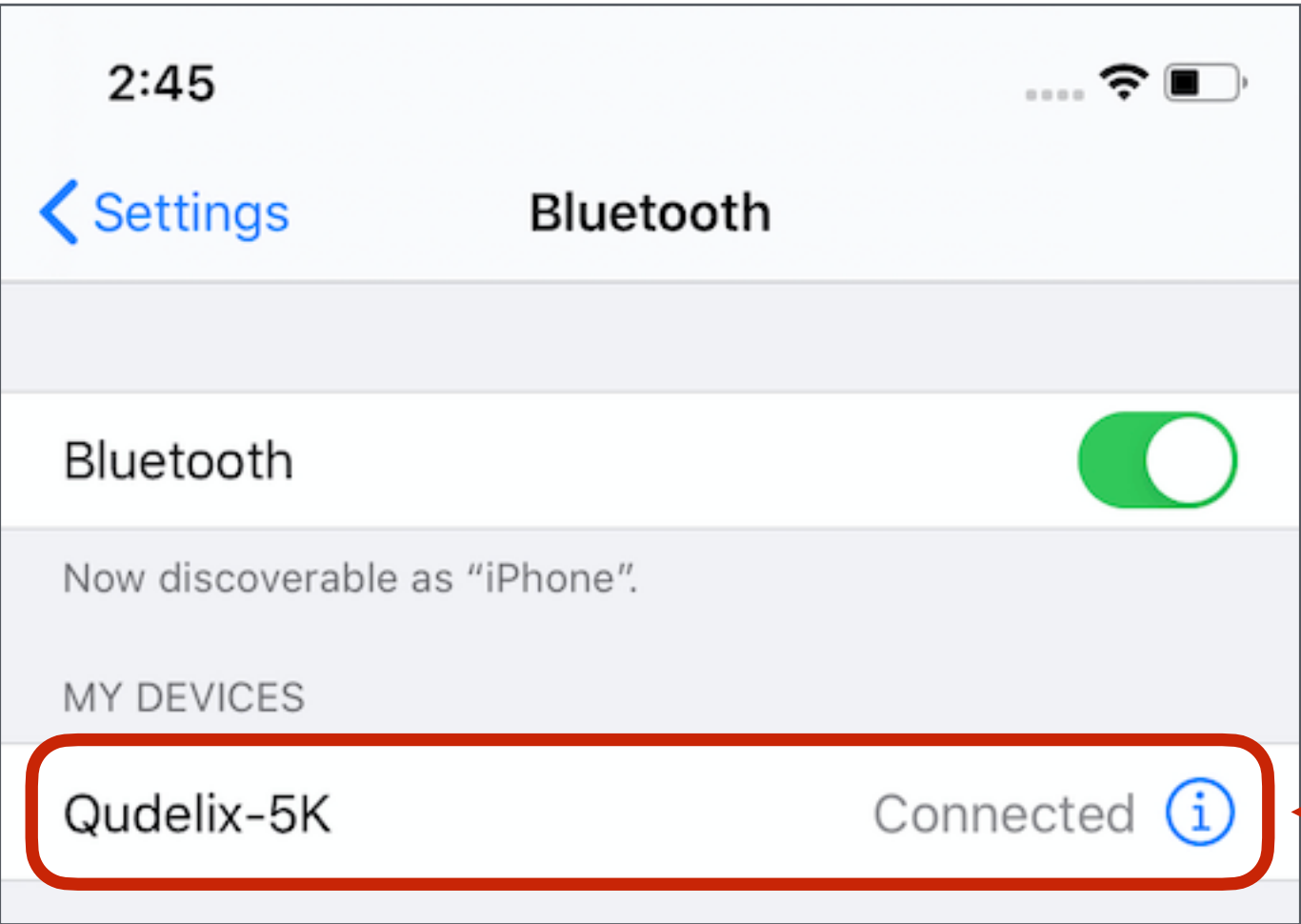
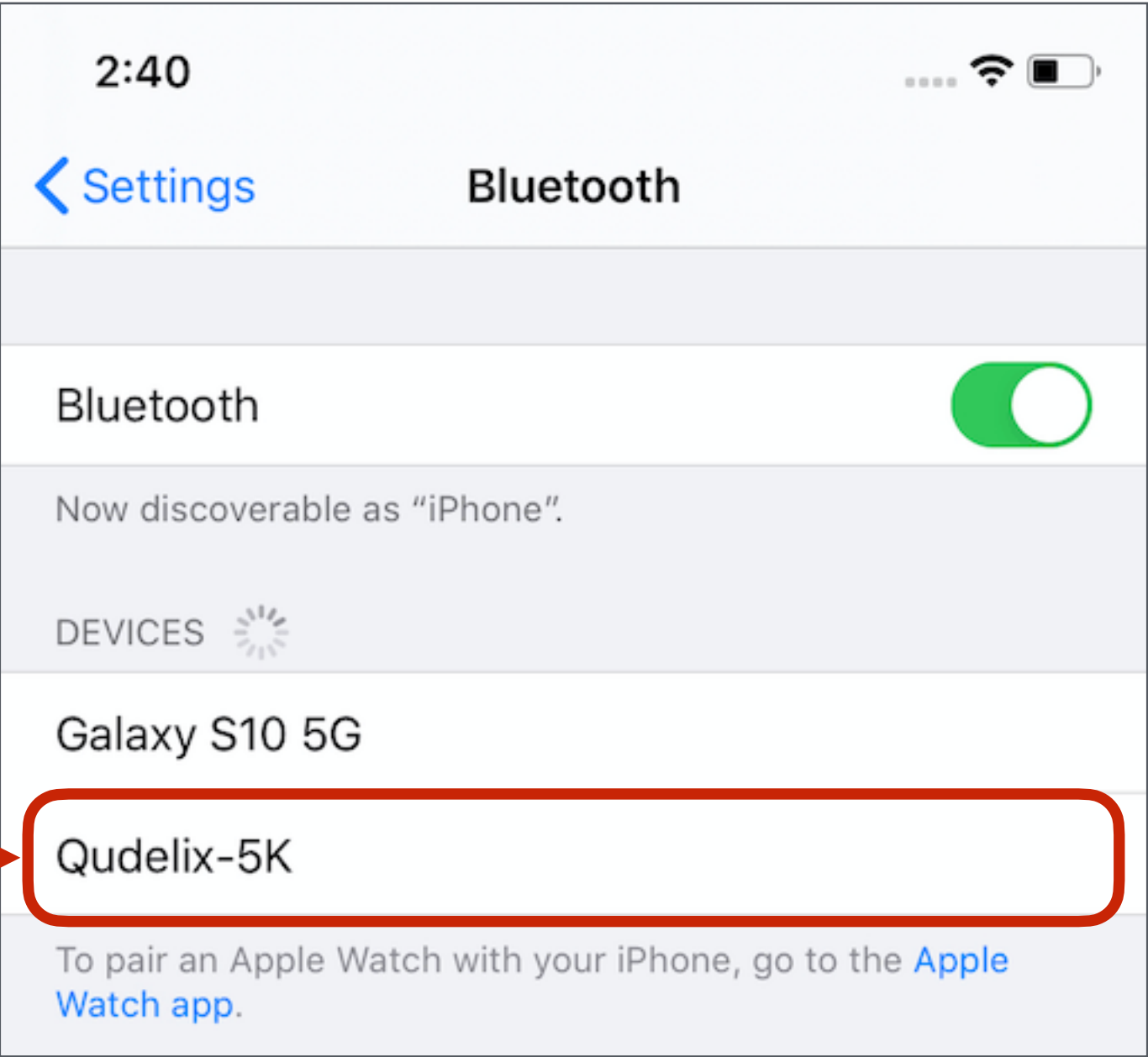
3. Go to your iOS Bluetooth menu

- Scan Bluetooth devices.
- Select Qudelix-5K in the Available devices list.
- If Qudelix-5K does not appear, please toggle the Bluetooth button OFF and ON, then try scanning again.

4. Paired

- Once paired successfully, you will see Qudelix-5K Connected ⓘ in the MY DEVICES list.

Scan and select



Audio/Call
Connected with ⓘ



Qudelix 5K Bluetooth Audio Connection

Pairing with iOS

- **Important**

- Displaying as Connected ⓘ means the device is recognized as an audio device for Bluetooth audio and phone calls. If it only shows as Connected (without ⓘ), it indicates that the Qudelix iOS app is connected to the device via a BLE (Bluetooth Low Energy) link.
 - Connected ⓘ - Bluetooth Audio / Phone Call Connection
 - Connected - BLE Connection for the Qudelix iOS App

- **iPhone Bluetooth Scanning Issue**

- In most cases where this occurs, iPhones may temporarily fail to scan for or discover nearby Bluetooth devices. This is a known iPhone/iOS issue and is not specific to Qudelix. If the Qudelix device does not appear during Bluetooth scanning, turn Bluetooth OFF on your iPhone, wait a few seconds, and then turn it back ON. This refreshes the Bluetooth scanning process and usually resolves the issue.

Mobile App - iOS

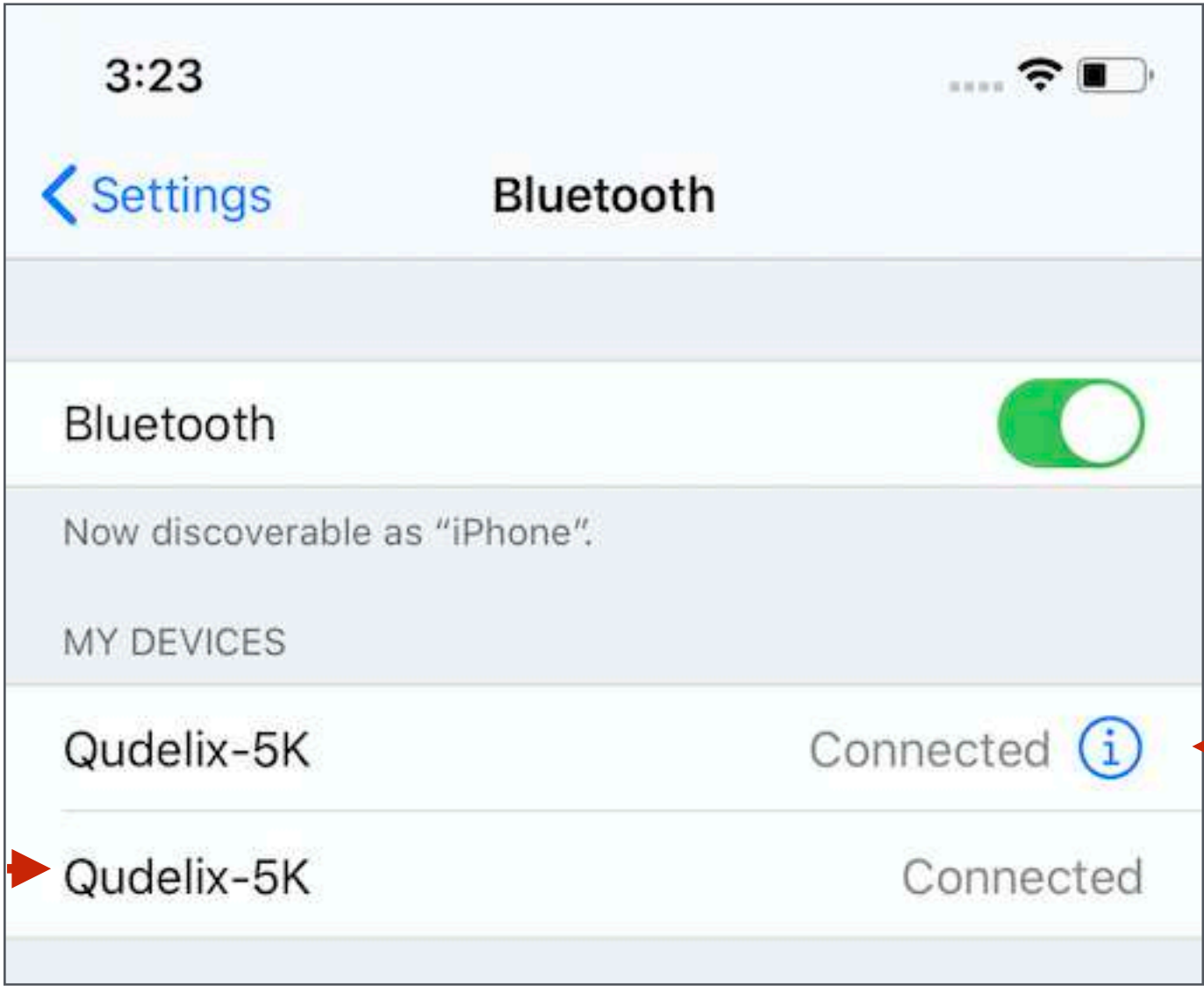


- Install iOS Mobile App
 - <https://apps.apple.com/kr/app/qudelix/id1515641059?l=en>



- Bluetooth audio streaming works independently of the Qudelix App. You can get audio streaming over Bluetooth without Qudelix App. However, we strongly recommend installing and using the app to fully experience the feature-rich nature of Qudelix-5K.
- The 5K supports an app connection with only one device at a time.
- iOS App uses the BLE (Bluetooth Low Energy) link.
 - 'Connected ⓘ' represents the Audio/Call connection.
 - 'Connected' represents the app connection over BLE.

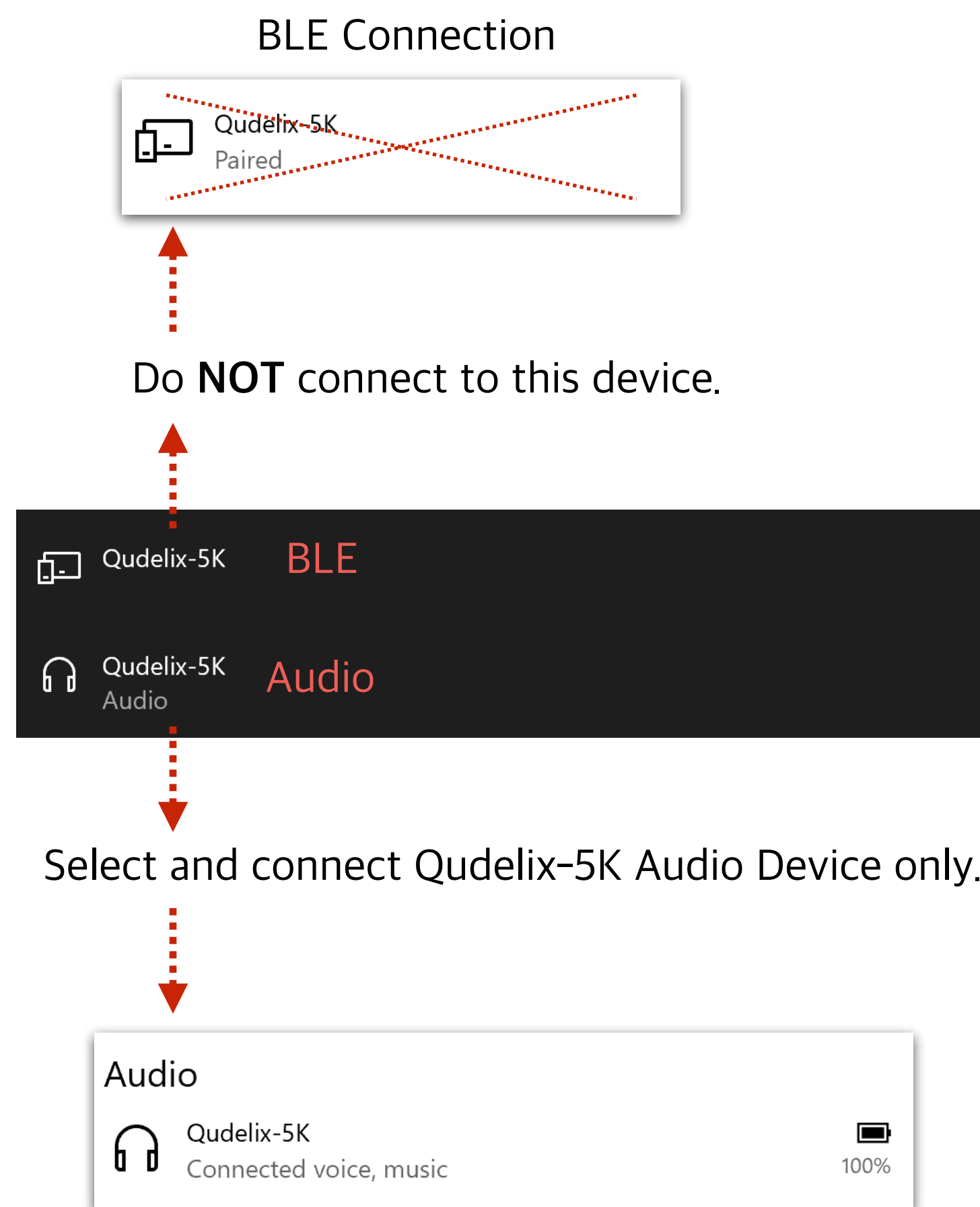
BLE App
Connection



Audio
Connection

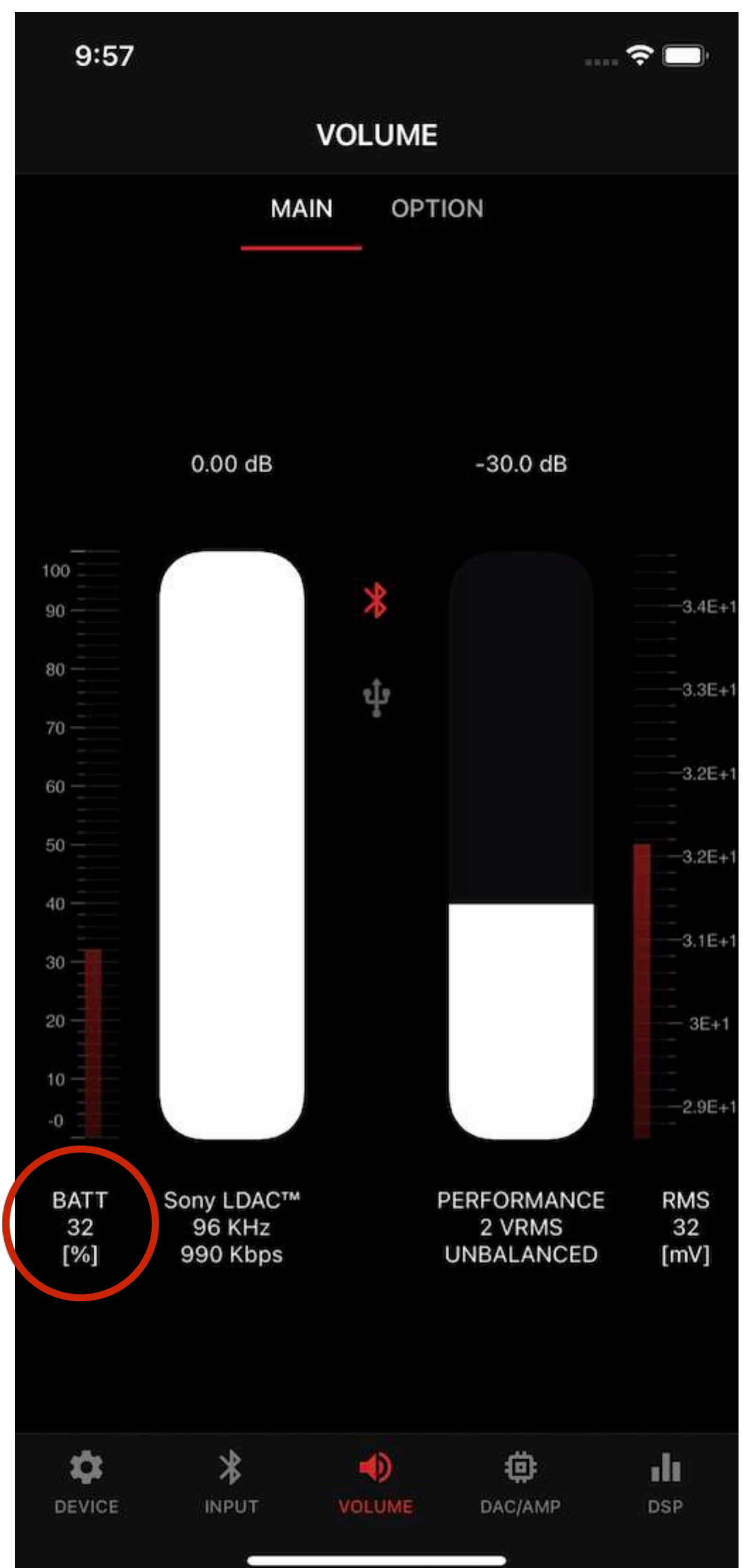
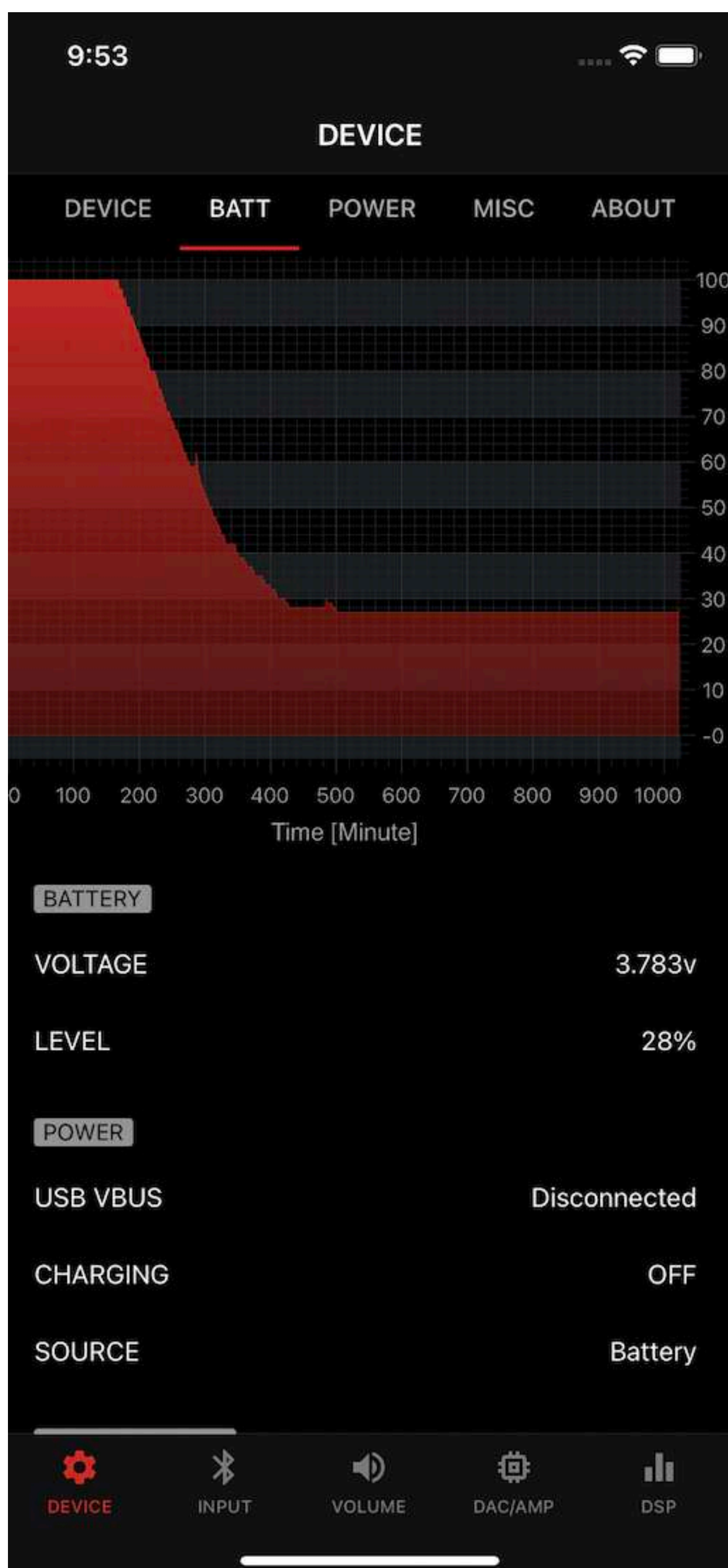
Pairing with Windows

- Do not make the BLE connection when pairing the 5K with a Windows PC.
- When searching for nearby Bluetooth devices in the Windows Bluetooth menu, you will see a Qudelix-5K Audio and a Qudelix-5K.
 - Qudelix-5K Audio (Headset icon) – for Voice Call & Music
 - Qudelix-5K (BLE) – for iOS/Android app connection
- Make the connection to Qudelix-5K Audio (Headset icon) only.
- Connecting the Qudelix-5K (BLE) will cause an iOS/Android app connection failure.



Battery

- The app displays the current battery and charger status.
 - Battery Log Graph
 - Instantaneous battery level
- The battery voltage is just for reference.
- The 5K built-in charger will limit the maximum charging current to 200mA.
- 500mA Battery / 2H 30MIN charging time from empty to full



Battery - Safety Warning

- Do not disassemble, puncture, crush, heat, or burn the device.
- Do not expose the device to excessive heat, direct sunlight, or fire.
- Do not leave the device inside a vehicle exposed to direct sunlight or high temperatures for extended periods. Excessive heat may damage the battery and create a risk of fire or explosion.
- Use only certified USB power sources and cables.
- Stop using the device immediately if it becomes unusually hot, swollen, or damaged.
- Keep the device away from water and high humidity.
- Do not attempt to replace or modify the internal battery.
- Improper use may result in fire, electric shock, injury, or explosion.



Battery Time

- Please note that the battery times may differ across each use case.
 - The battery time varies across the codec, sample rate, power profile, and output mode.
 - The sensitivity of the earphones/headphones and the output volume level also can make the battery time longer or shorter.

AAC 44.1 kHz

	Standard Mode		Performance Mode	
	Current Avg. [mA]	Time [Hour]	Current Avg. [mA]	Time [Hour]
Balanced 4V Dual DAC	51.30	8.77	66.80	7.49
Balanced 2V Dual DAC	40.00	12.50	51.90	9.63
Unbalanced 2V Single DAC	30.60	16.34	38.00	13.16
Unbalanced 1V Single DAC	25.00	20.00	30.40	16.45

LDAC 44.1 kHz @ 909 kbps

	Standard Mode		Performance Mode	
	Current Avg. [mA]	Time [Hour]	Current Avg. [mA]	Time [Hour]
Balanced 4V Dual DAC	55.00	9.09	70.20	7.12
Balanced 2V Dual DAC	42.70	11.71	55.40	9.03
Unbalanced 2V Single DAC	33.00	15.15	40.50	12.35
Unbalanced 1V Single DAC	27.20	18.38	32.80	15.24

LDAC 96 kHz @ 990 kbps

	Standard Mode		Performance Mode	
	Current Avg. [mA]	Time [Hour]	Current Avg. [mA]	Time [Hour]
Balanced 4V Dual DAC	59.20	8.45	74.50	6.71
Balanced 2V Dual DAC	46.90	10.66	59.70	8.38
Unbalanced 2V Single DAC	35.40	14.12	43.10	11.60
Unbalanced 1V Single DAC	29.80	16.78	36.10	13.85

ES9219 DAC/AMP - Amplifier Gain

- **User-selectable amp gain mode**
 - Normal 1 V RMS
 - High 2 V RMS
 - Both provide the same sound performance.
- **Normal 1 V RMS mode**
 - Recommended for the most IEMs
 - Longer battery time and the lower noise floor
- **High 2 V RMS mode**
 - 6dB higher maximum output level
 - Recommended for headphones
- **Auto Volume Correction**
 - The firmware adjusts the volume and keeps the current output level when switching the gain mode.
 - You will experience no difference in loudness when switching the mode to 2 V RMS. But you will have the max volume extended by +6dB.
- The gain mode switching requires a couple of seconds of audio stabilizing time. You may experience some audio artifacts during the transition period.



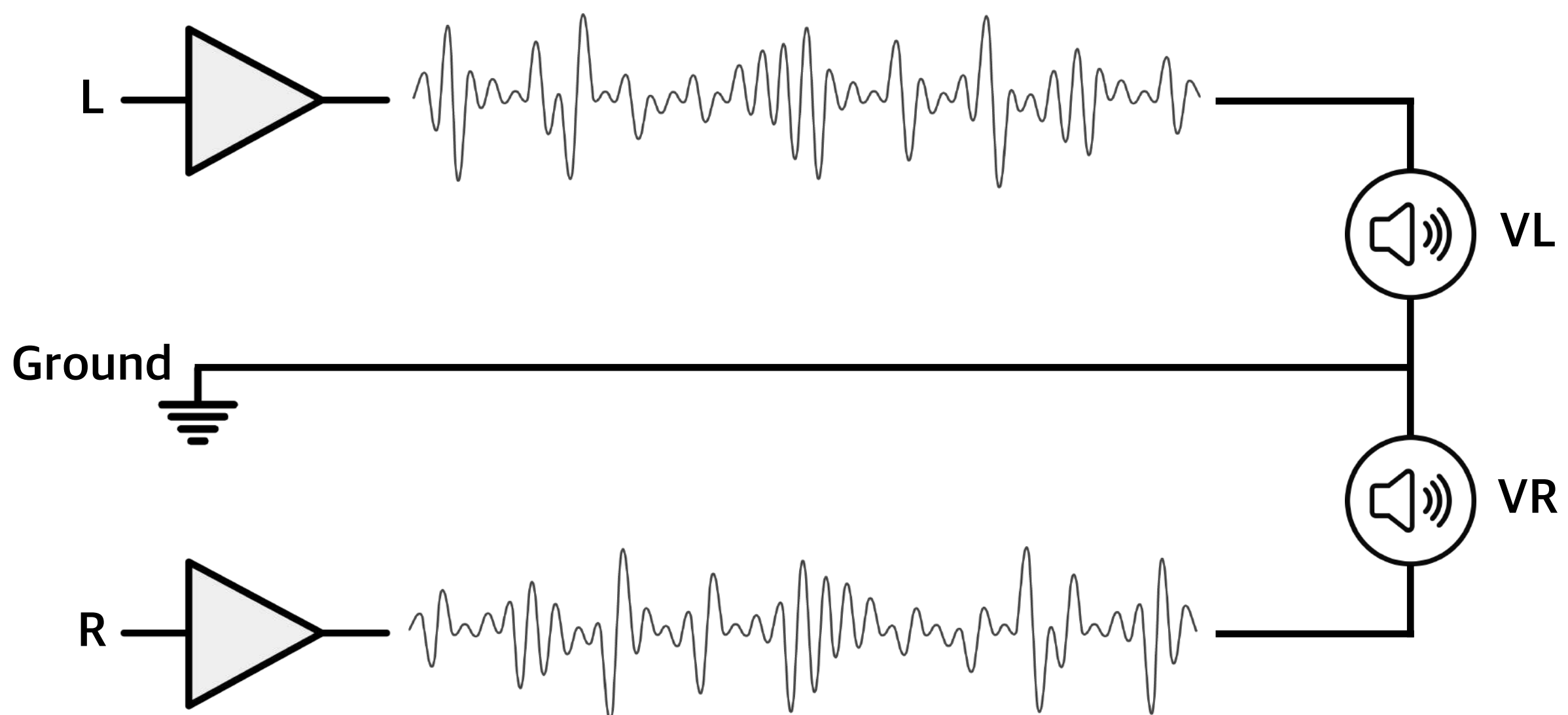
ES9219 - DAC Operating Profile

- Qudelix's proprietary software drives ES9219 DACs in two ways to trade off the battery time and sound quality.
- **Standard Profile (ECO mode)**
 - Standard DAC Operating Clock
 - Standard Output Bias Current
 - Best Efficiency
 - The best trade-off for battery time and sound quality
 - Suitable for mobile and outdoor use cases
- **Performance Profile (SPORT mode)**
 - Highest DAC Operating Clock
 - High Output Bias Current
 - Best Performance
 - Prioritizes audio performance over battery life
- Supplying external USB power automatically overrides the current profile and switches the device to PERFORMANCE mode.



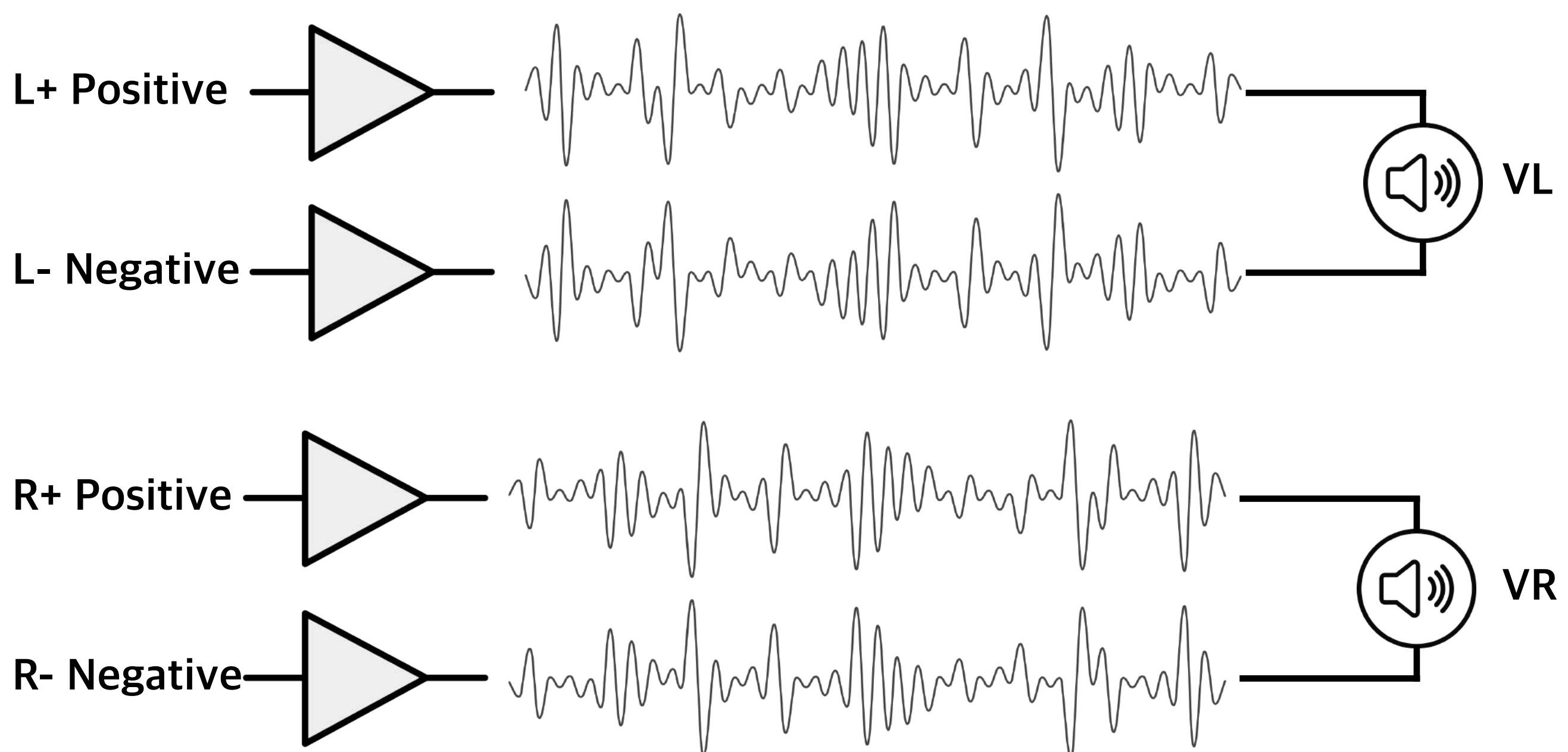
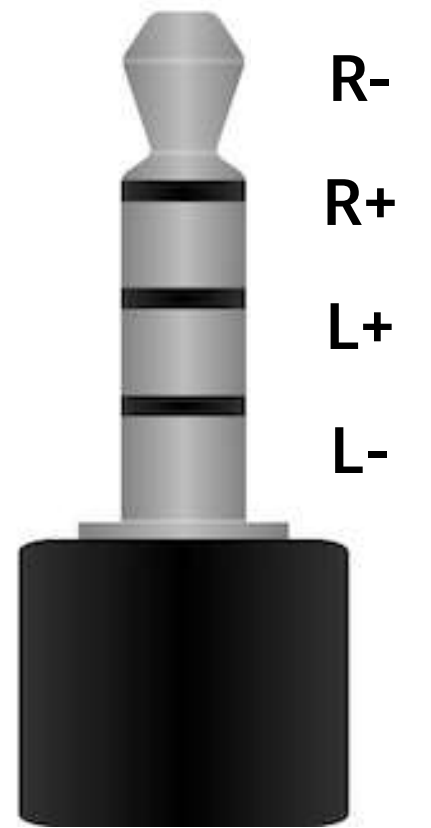
UNBALANCED (Single-Ended) 3.5mm

- Works with Single ES9219
 - Max. 2 V RMS output
 - 80 mW per channel
- 3-Wire Interface
 - 2 ch DAC/AMP for Stereo Audio
 - 1-pin Ground
- Common Ground
 - Ground Pin goes to both Speaker-L and Speaker-R
 - Ground ripple (noise) affects both L/R channel
- Both AMP_NOISE and GROUND_NOISE affect the Sound Quality
 - $VL = L_AUDIO + AMP_NOISE + GROUND_NOISE$
 - $VR = R_AUDIO + AMP_NOISE + GROUND_NOISE$



BALANCED (Full-Differential) 2.5mm

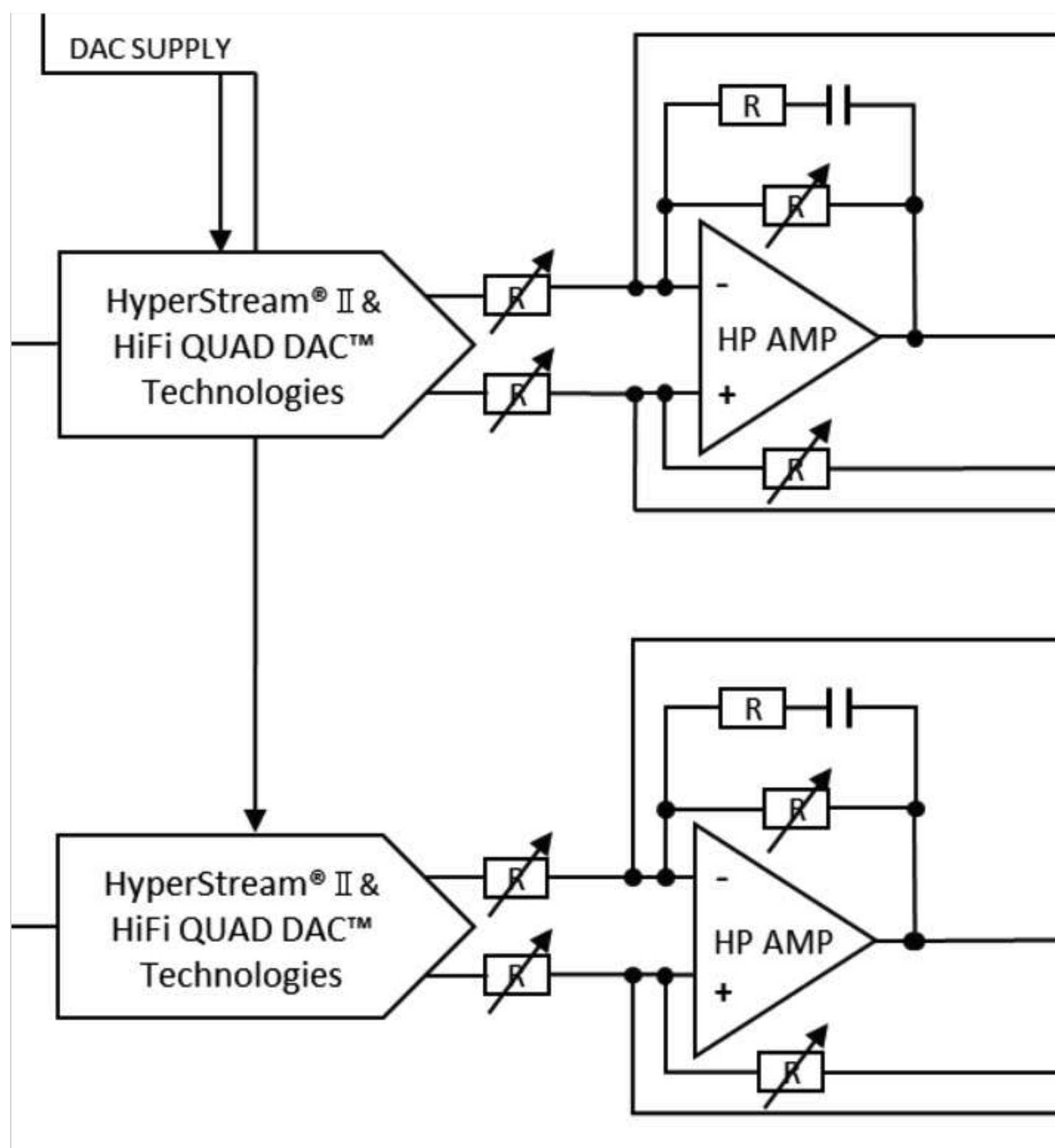
- Works with Dual ES9219
 - Max. 4 V RMS output
 - 240 mW per channel
- 4-Wire Interface
 - 4ch DAC/AMP fully wired to the stereo speaker terminals
 - Original Signal (L+/R+) and Inverted(Negative) Signal (L-/R-)
- Free from Ground Noise
- Common Mode Noise Reduction
 - Common AMP_NOISE is canceled out easily
 - $VL = (LP_AUDIO + AMP_NOISE) - (LN_AUDIO + AMP_NOISE) = LP_AUDIO - LN_AUDIO = 2L_AUDIO$
 - $VR = (RP_AUDIO + AMP_NOISE) - (RN_AUDIO + AMP_NOISE) = RP_AUDIO - RN_AUDIO = 2R_AUDIO$
- Relatively better Sound Quality
 - Better Linearity
 - Better Channel Separation
 - 2x more power and driving capability



*With any given hardware device supporting UNBAL and BAL, BAL typically provides superior performance compared to UNBAL, and we strongly recommend using BAL whenever possible.

ES9219 Built-In PGA (Programmable Gain Amp)

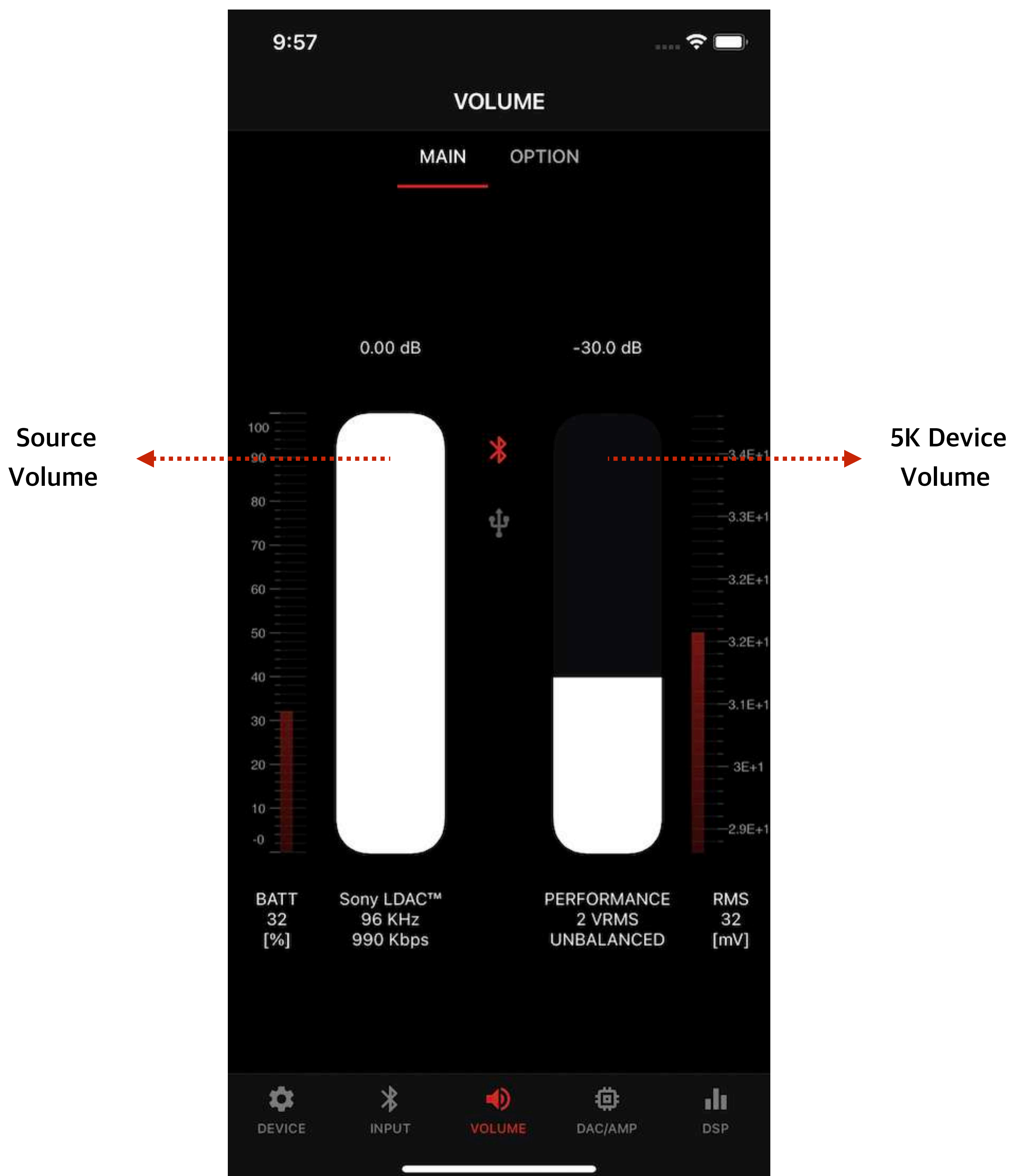
- Analog Volume Control
 - Integrated PGA(Programmable Gain Amp)
 - 0dB ~ -24dB Range
 - 1dB Step
 - Provides the best sound quality at any volume level
 - Noise level gets lowered as the volume level gets reduced.
- 5K manages the output level with the ES9219 built-in PGA, providing optimal sound quality at any given volume level.



ES9219 DAC & HPAMP
with PGA(Programmable Gain Amplifier)

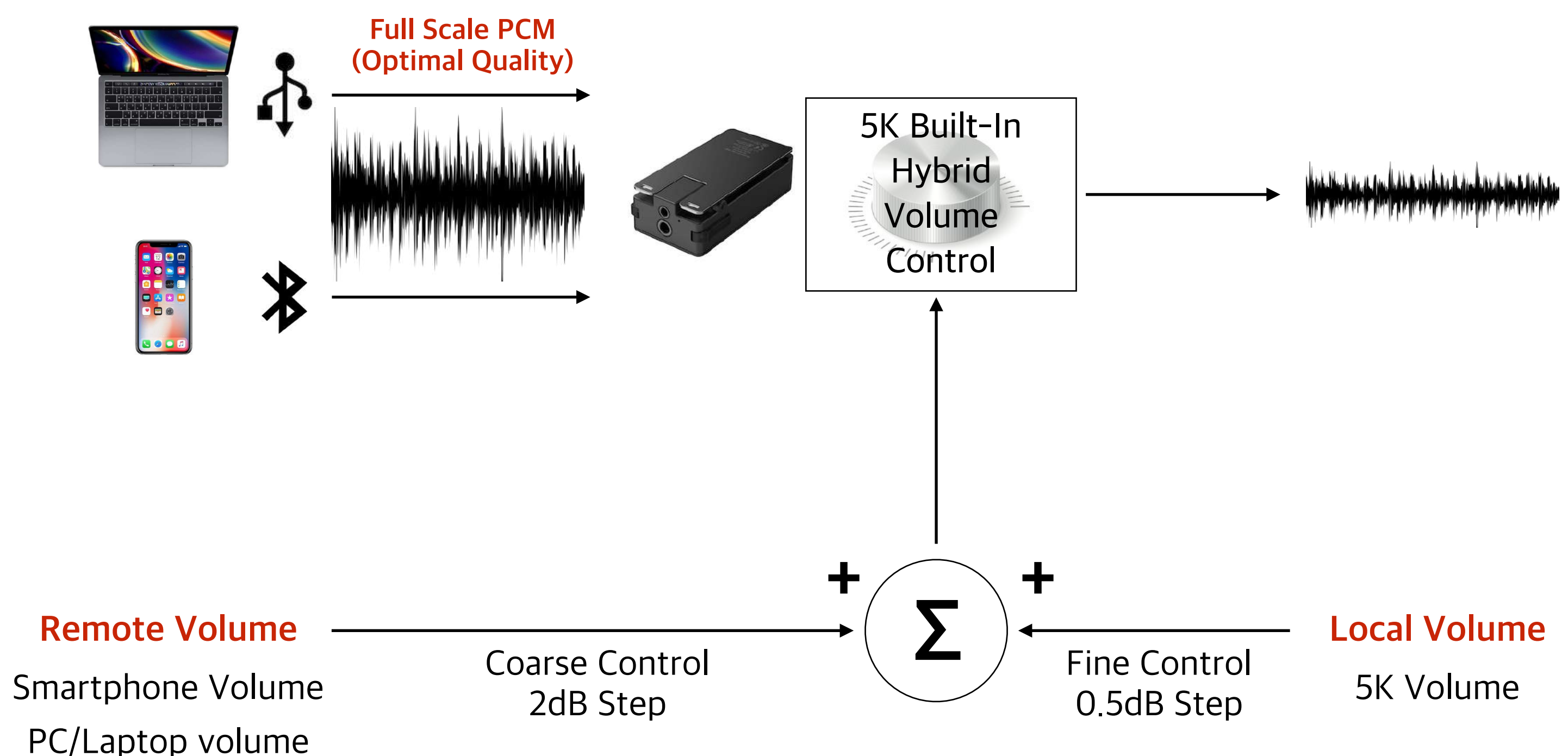
Volume

- 5K offers 2-way associative absolute volume control for optimal sound quality and always keeps and preserves the audio stream at full scale over Bluetooth or USB.
- Use whichever method is most comfortable for you.
 - Use the smartphone volume for coarse adjustment.
 - Use the 5K volume for fine adjustment.
 - They both provide the same sound quality.



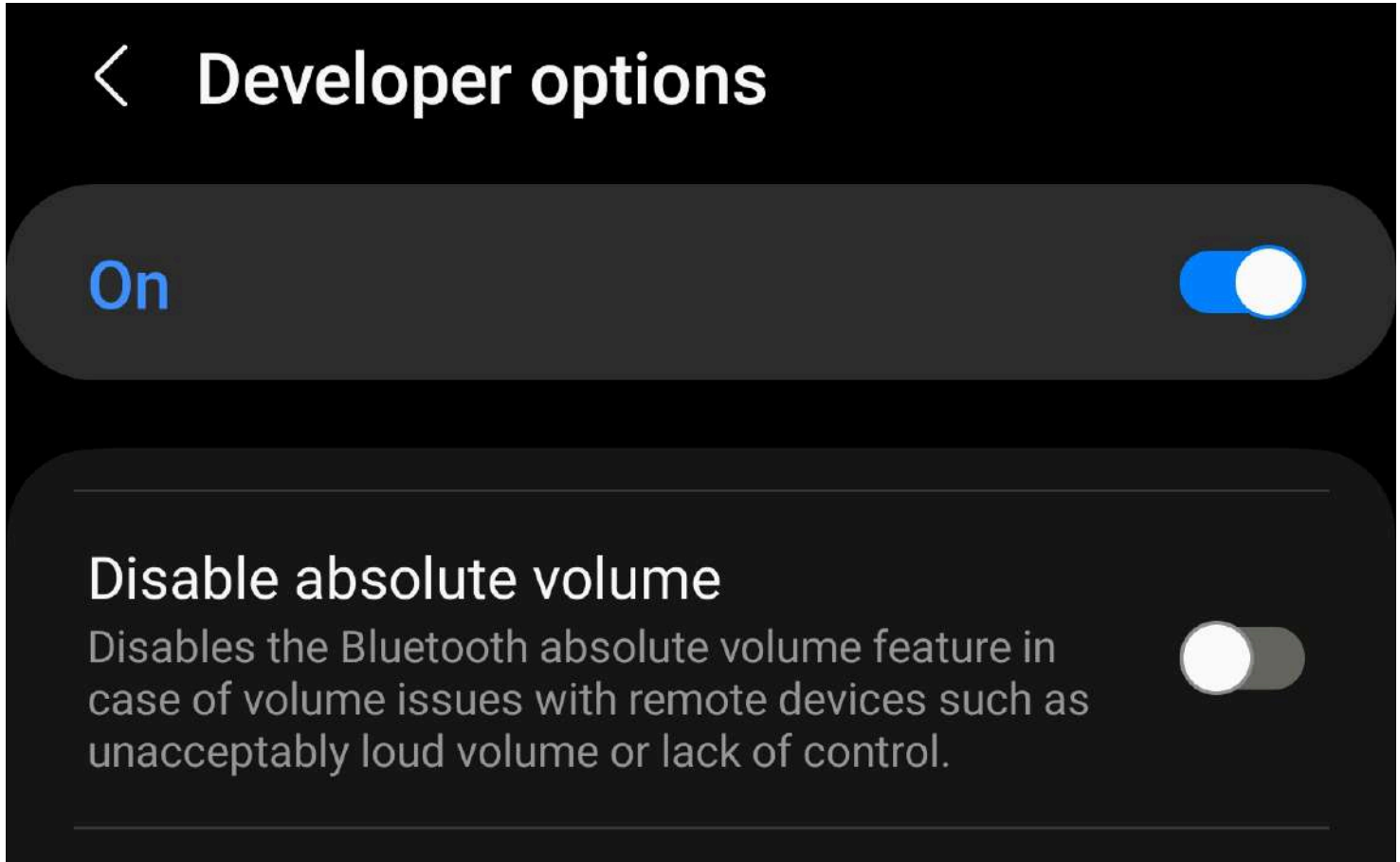
2-way Associative Absolute Volume

- Qudelix's proprietary 2-way associative absolute volume control provides optimal sound quality at any source or device volume level. With this scheme, the source device always delivers the audio stream at maximum precision, preserving the original signal quality.
- You can control the volume using either the smartphone buttons or the Qudelix-5K buttons without any loss in sound quality. It always preserves the best sound quality over Bluetooth or USB.
- **Smartphone, PC/Laptop volume control**
 - Remotely adjusts 5K built in Volume Controller
 - 2dB step Coarse Control
- **5K Device Volume Control**
 - 0.5dB step Fine Control
- **Optimal sound quality for any volume settings**
 - Any smartphone's or 5K's volume setting provides the best sound quality preserving maximum audio precision.



2-way Associative Absolute Volume

- Android provides a Bluetooth setting called “Disable absolute volume.” As shown below, this option should remain OFF.

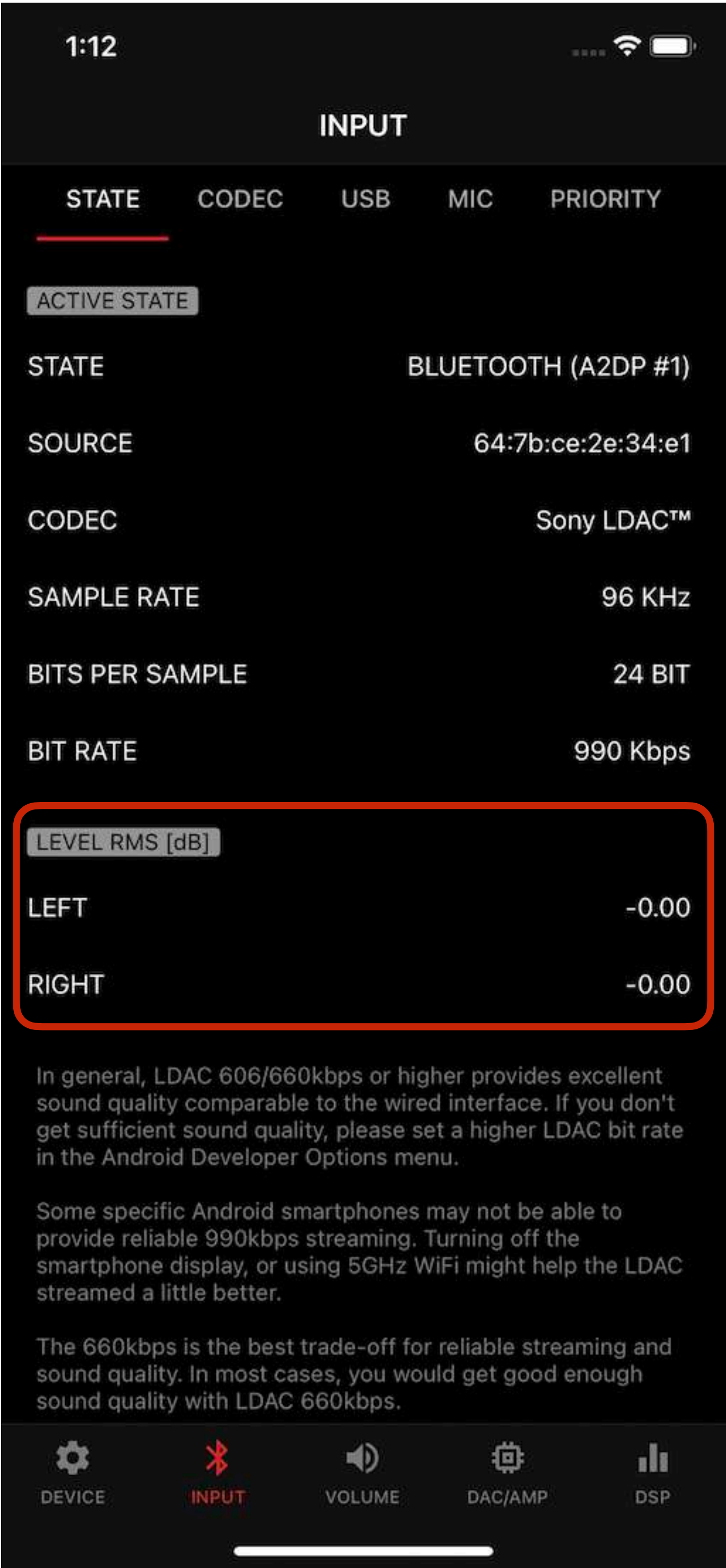


- How to check if the ABS volume is working
 - When you adjust the source volume (i.e., smartphone or pc volume), the source device sends the volume command to the 5K. And the 5K will blink the Blue LED once it receives the volume command.
 - If Absolute Volume is disabled, the source scales down the PCM audio and sends no command to the 5K. Thus the 5K Blue LED will not blink.
 - To confirm the ABS volume is working, check if the 5K Blue LED blinks upon the source device volume adjustment.
- NOTE: Android USB Audio doesn't support Absolute Volume Control.

Source	ABS Volume Support
Windows/macOS/Linux USB	O
iOS USB	O
Android USB	X
Bluetooth	O

Volume & Input RMS

- The Level RMS will display the instantaneous average of the incoming audio for every 1 second.
 - The level will change continuously as a music track goes on.
 - Or, if you play a static test tone or noise with a constant level, they will keep displaying a certain level.
- Monitoring the incoming RMS level at the 5K is one way to check if Absolute Volume Control is working. Without the ABS volume control, the RMS level will decrease as the source device volume decreases.



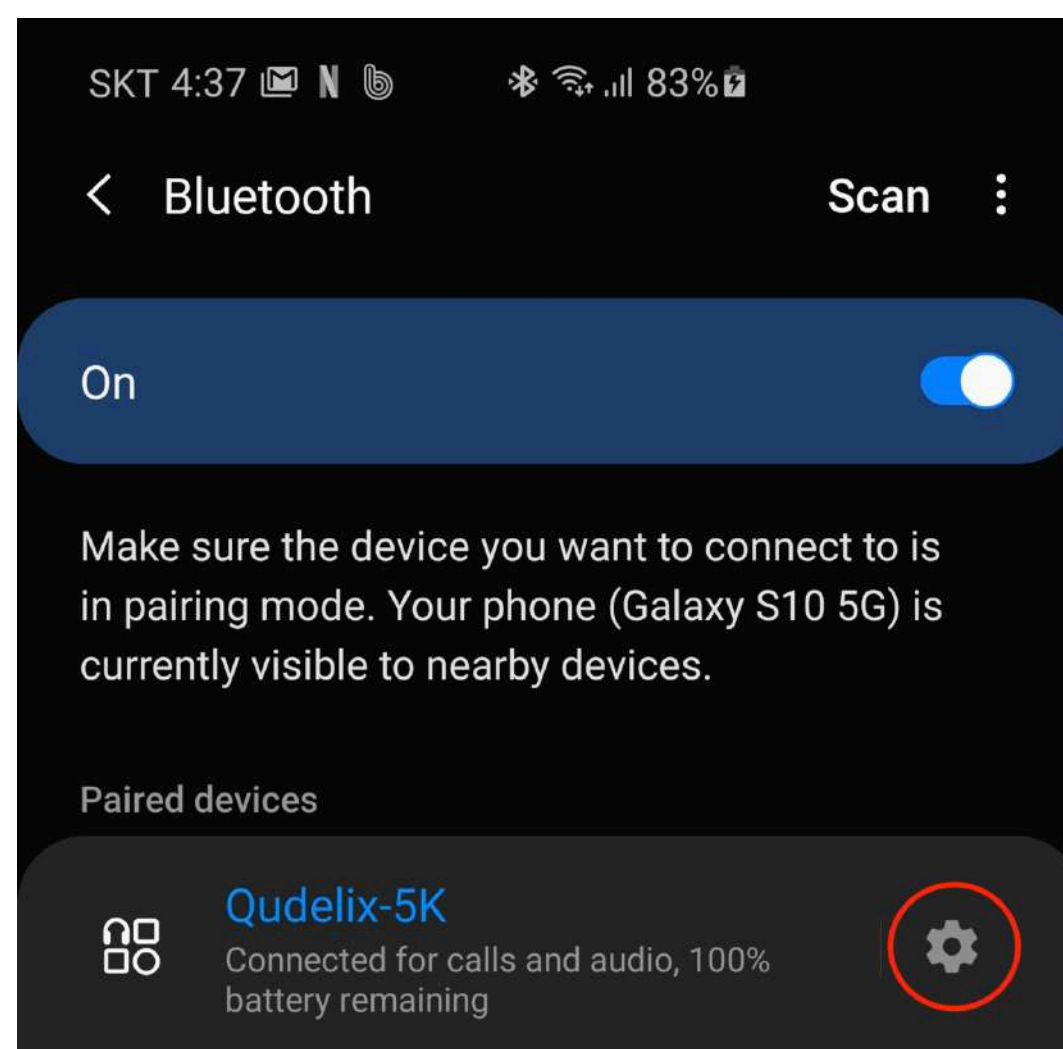
Bluetooth Codec - LDAC

- **SONY LDAC**

- Available with Android 8.0 or higher.
- LDAC 606/660 kbps or higher generally provides excellent sound quality comparable to the wired interface. If you don't get sufficient sound quality, please set a higher LDAC bit rate in the Android Developer Options menu.
- Some specific Android smartphones may not be able to provide reliable 990 kbps streaming. Turning off the smartphone display or using 5GHz WiFi might help the LDAC stream a bit better.
- The LDAC 660 kbps is the best trade-off for reliable streaming and sound quality. In most cases, you would get good enough sound quality with LDAC 660 kbps.

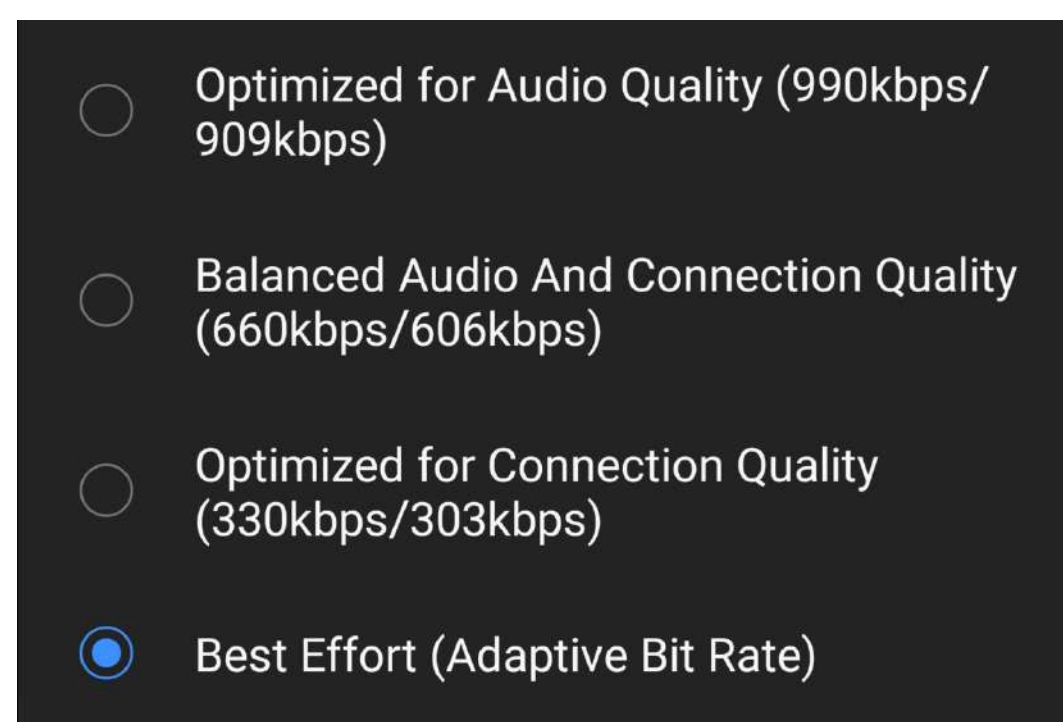
- **Enable SONY LDAC**

- Bluetooth Menu → Paired Devices → Qudelix-5K → device option → LDAC



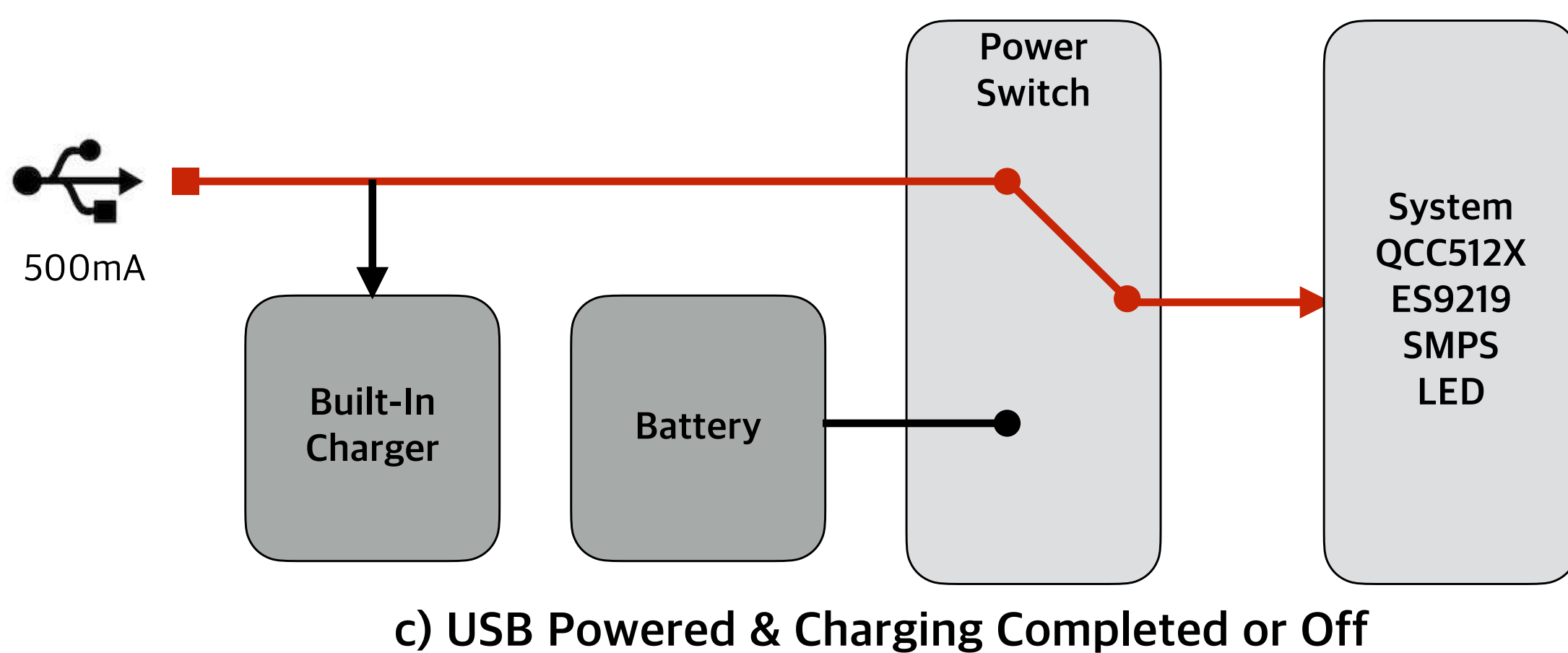
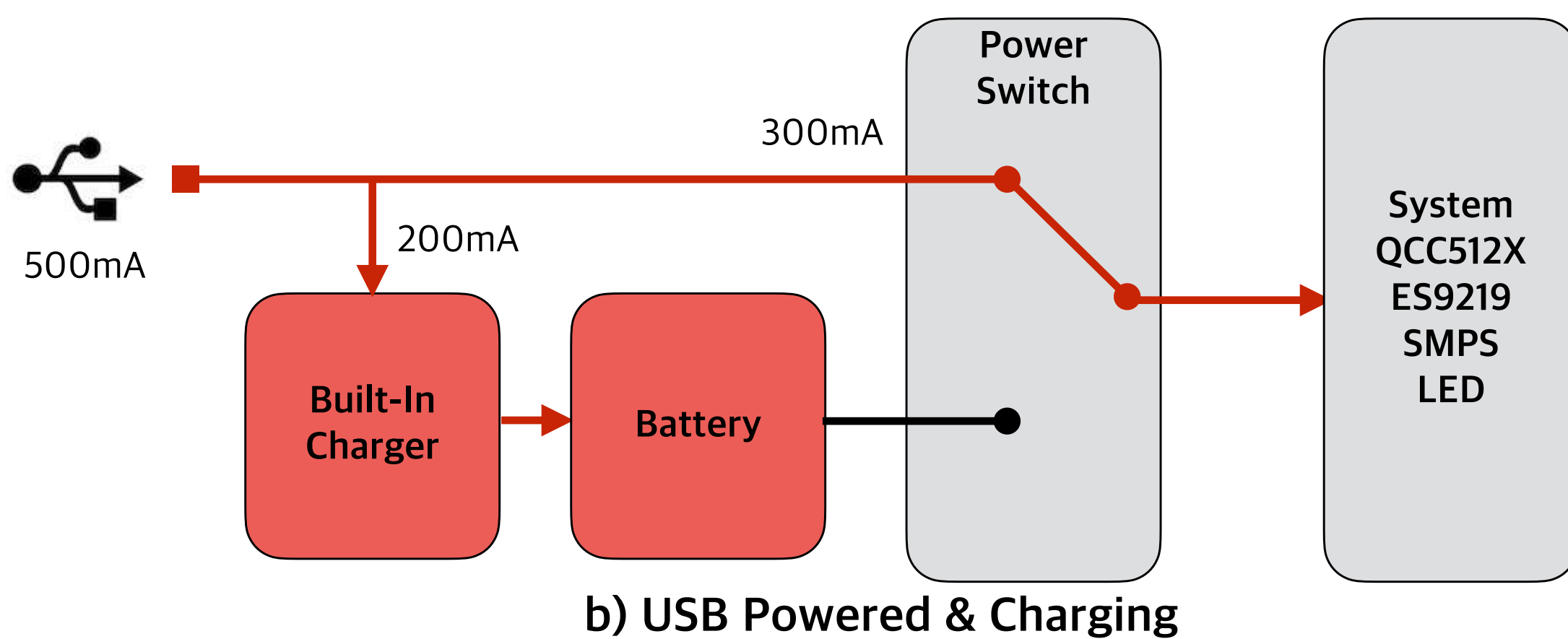
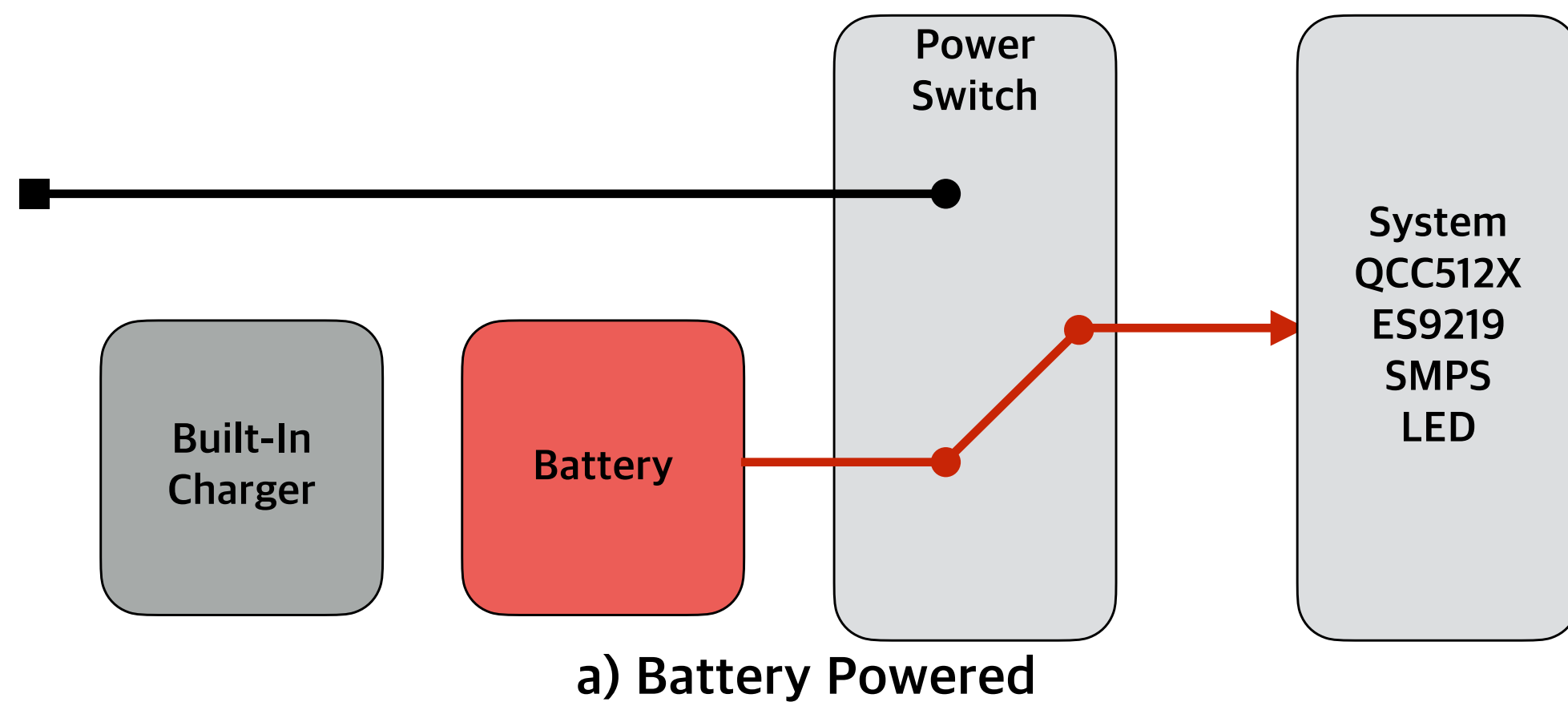
- **Setting LDAC Bitrate & Sample Rate**

- Enable Android Developer Option
 - <https://developer.android.com/studio/debug/dev-options?hl=en>
 - Please refer to the link above and enable the developer options menu.
- System Setting → Developer Option → LDAC Quality



Power & Battery Management

- 5K has three power management modes, as shown in the block diagrams below.
 - Battery Powered
 - USB Powered & Charging ON
 - USB Powered & Charging Off

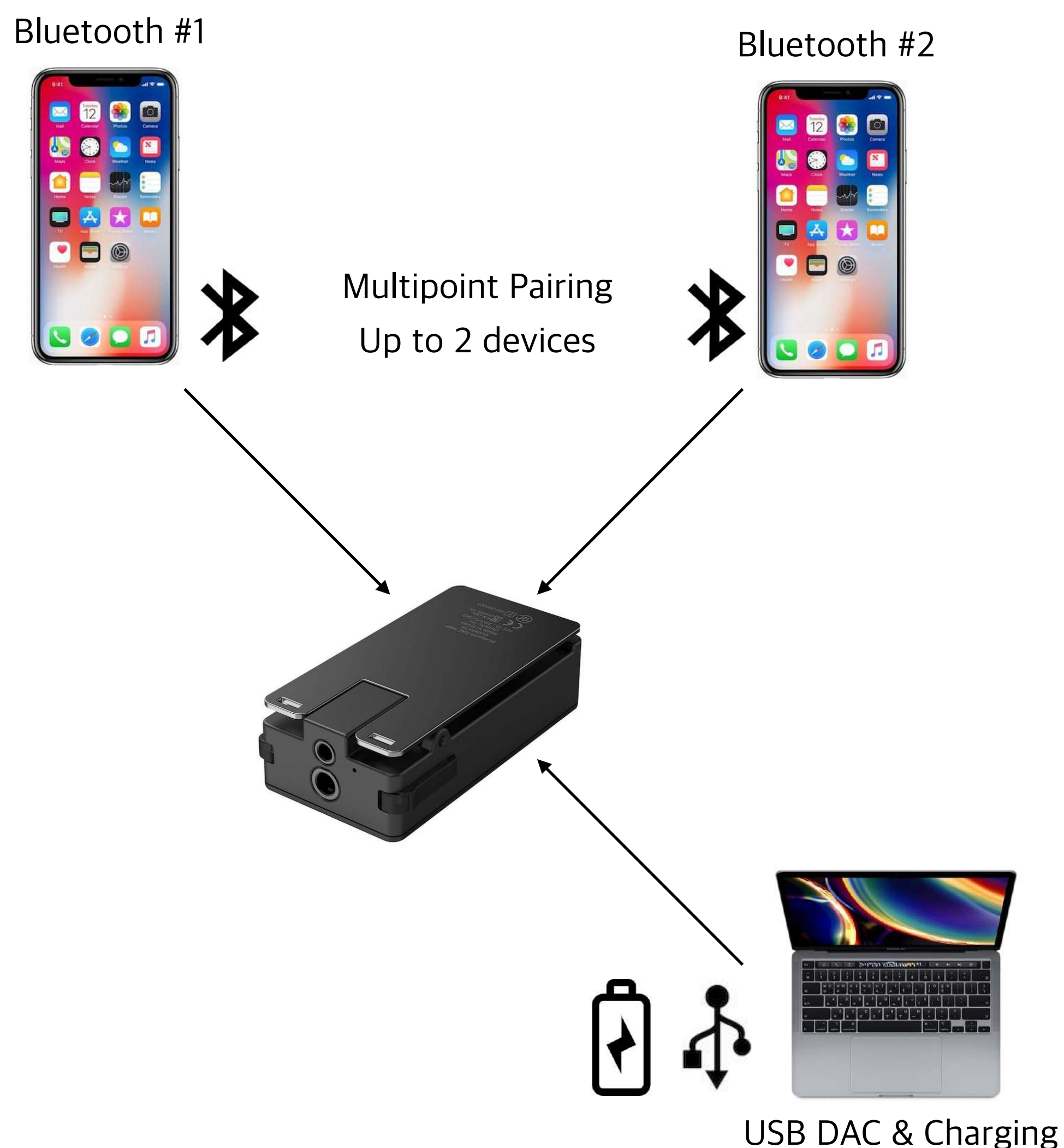


Power & Battery Management

- **Battery Powered Mode**
 - 5K works with the built-in battery.
 - It's the most typical use case for wireless streaming over Bluetooth.
- **USB Powered with charging on**
 - You can use 5K while charging the battery without any worries.
 - 5K always sources its operating power from the USB VBUS, if available.
 - For example, when you connect the 5K to a PC USB, smartphone USB, or USB charger, 5K automatically switches the power source from the battery to the USB through the built-in power switch.
 - The external USB VBUS provides 5K operating power and charges the built-in battery simultaneously.
 - It keeps the battery life longer and guarantees static battery charging time.
 - When the battery gets fully charged, 5K automatically turns off the built-in charger, cutting the current flow to the battery.
- *Some devices always draw operating power from the battery. In such cases, the battery may charge and discharge simultaneously, potentially reducing battery lifespan and generating additional heat.
- **USB Powered with charging off**
 - You can manually turn off the 5K built-in charger for mobile USB DAC use case.
 - iOS devices can supply output current to the accessory up to 150mA only.
 - If you connect the 5K to an iPhone over USB, you must turn off the 5K built-in charger. Otherwise, the 5K will draw too much power from the iPhone, and consequently, the iPhone will drop the connection with the accessory.

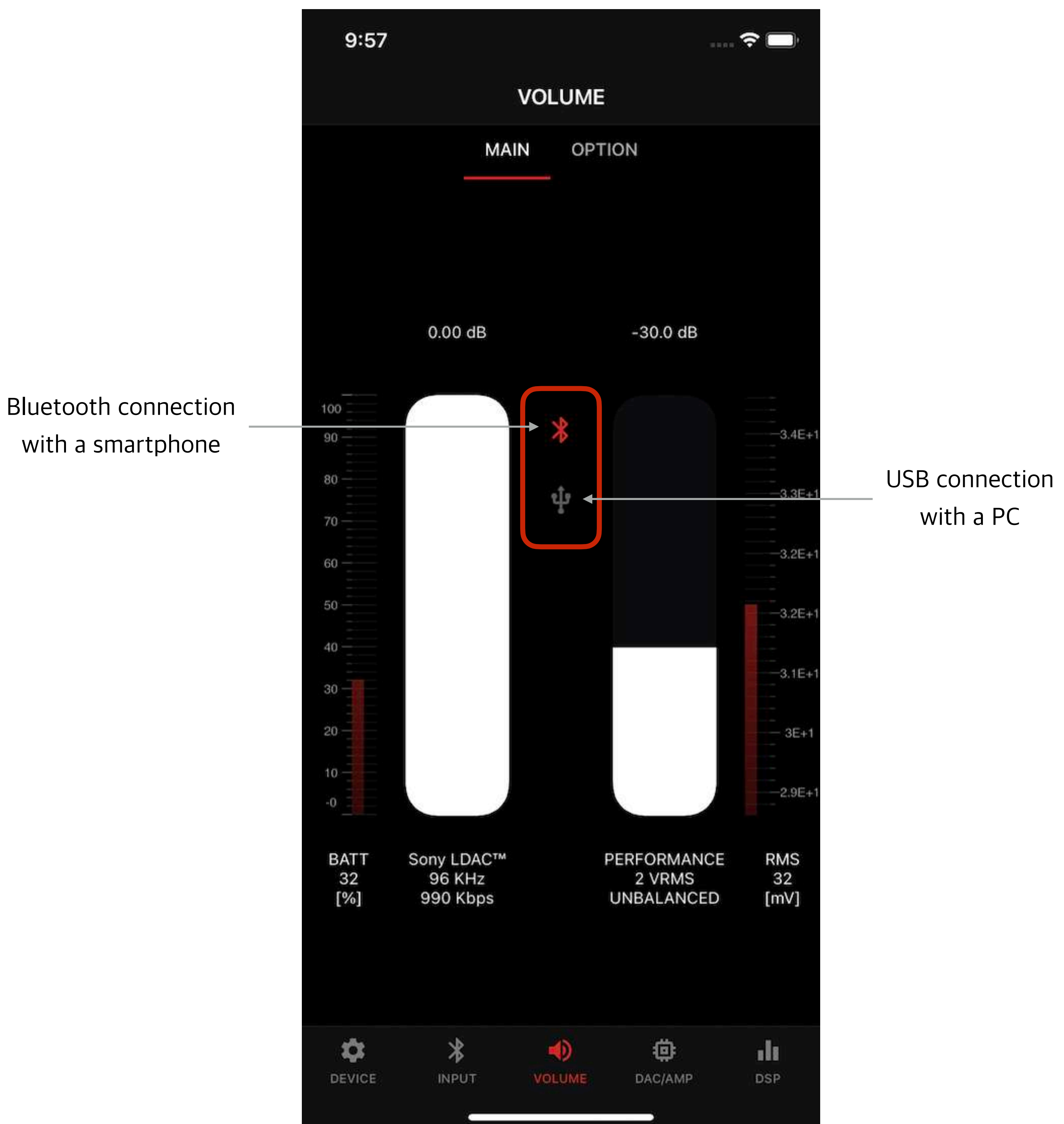
Connecting 5K to multiple source devices

- **Multi-Functional DAC**
 - 5K is a multi-functional DAC supporting Bluetooth A2DP and USB Audio Class.
 - As in the block diagram below, you can have multiple connections simultaneously with two Bluetooth Audio source devices and one USB Audio source device.
- **5K can keep the connections with multiple source devices, Bluetooth or USB, but it can stream audio from only one source device at a time.**
 - For example, a smartphone and a PC can stream audio to the 5K simultaneously over Bluetooth and USB, respectively. In this case, 5K needs to select one input source based on its priority policy.



Input Source Priority

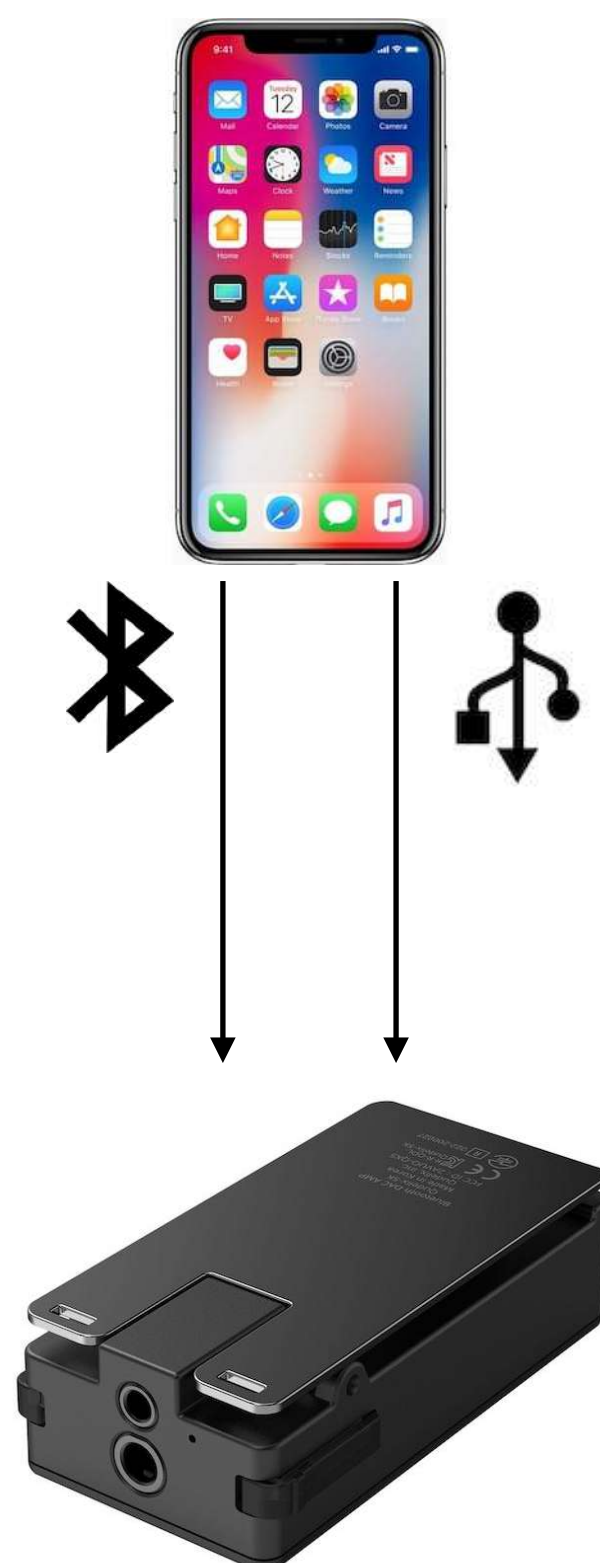
- **Multiple connections with multiple source devices**
 - 5K can stream audio from one source device at a time, thus, when multiple source devices stream audio to a 5K simultaneously, 5K selects one input.
- **Bluetooth and USB icon buttons**
 - They appear when 5K connects with each source device over the respective link.
 - The icon in Red represents the active audio source currently streaming.
 - The icons are buttons, and pressing an icon would change the input priority.



Multiple connections with a single source device

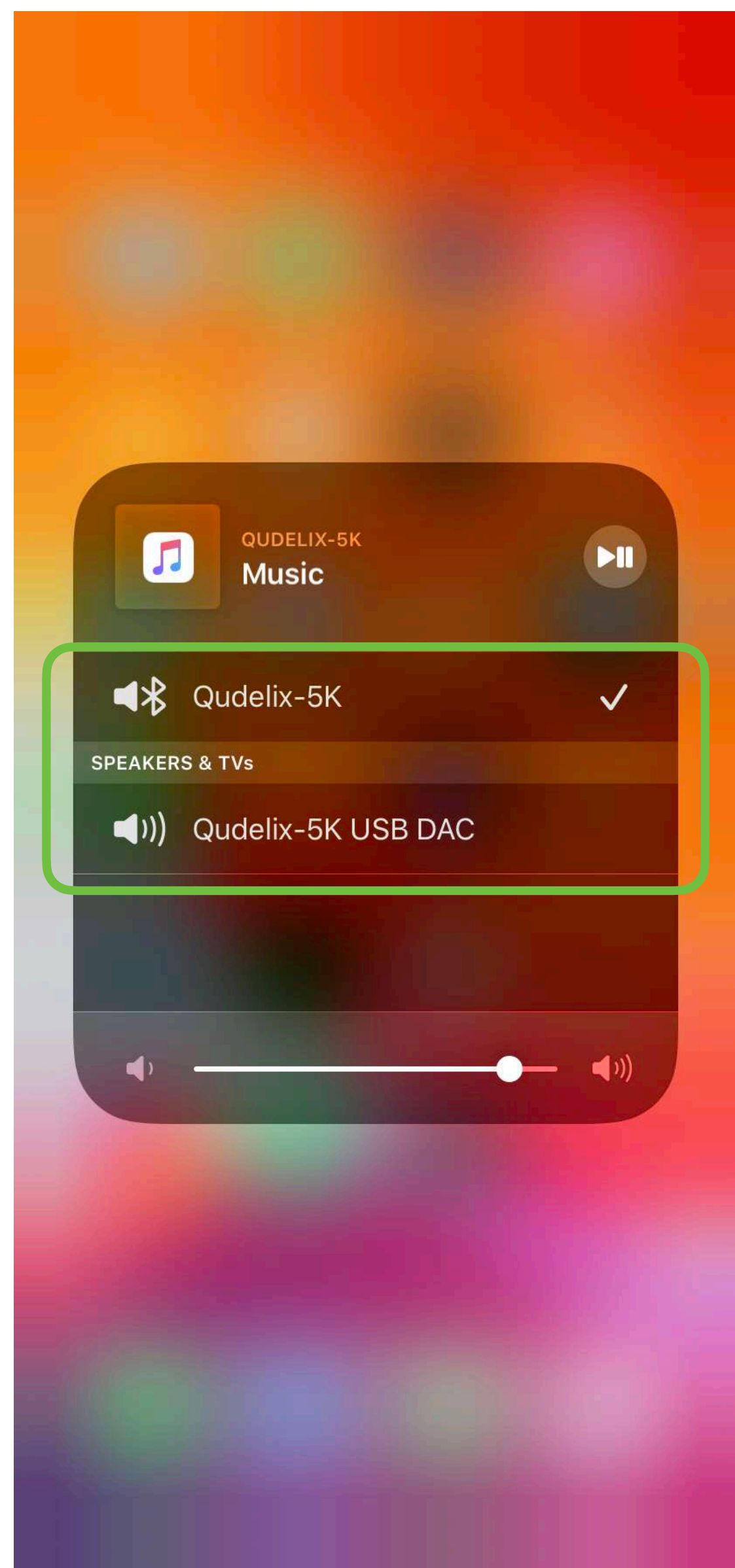
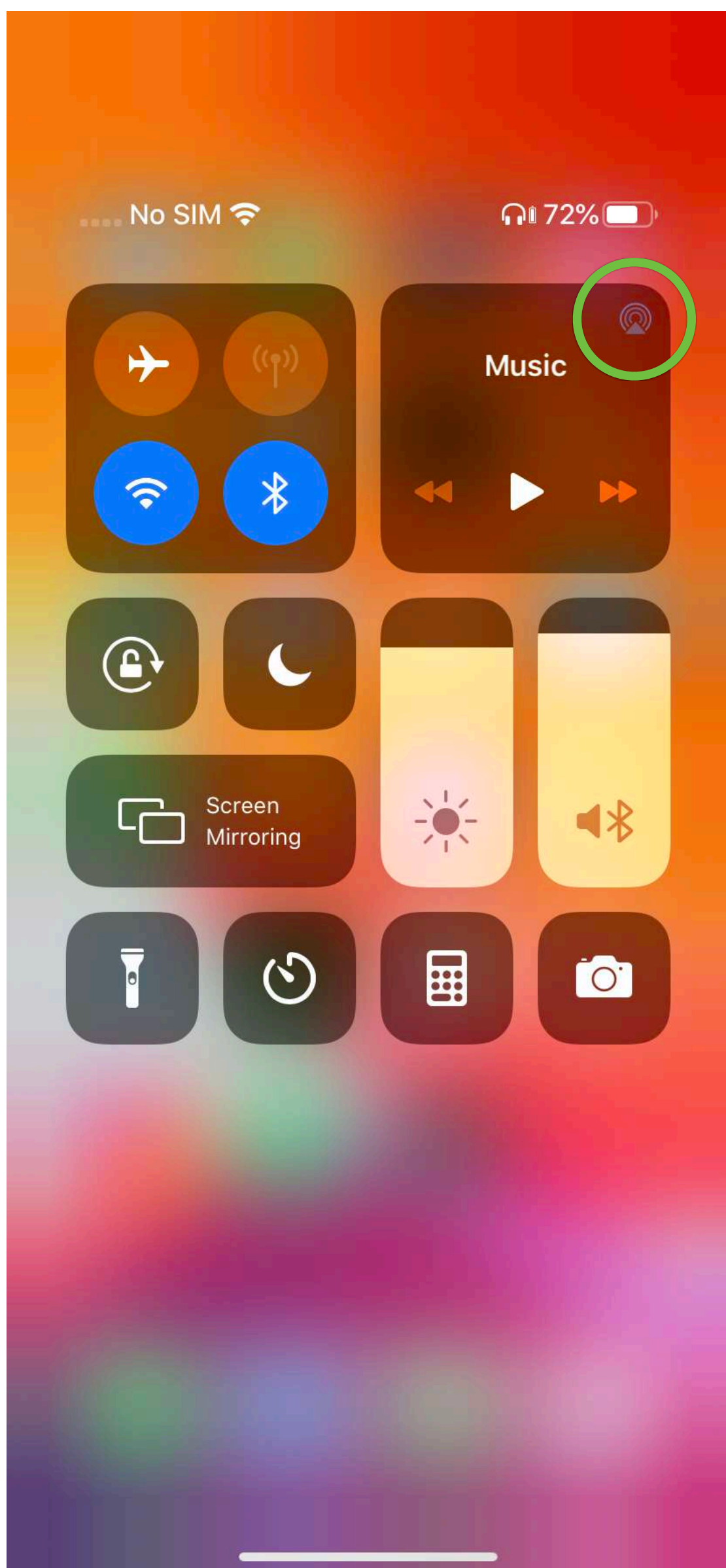
- **Multiple Audio Output Devices**

- You can connect a 5K to a smartphone with Bluetooth and USB. In this case, the smartphone has two output devices; one is 5K Bluetooth, and the other one is 5K USB.
 - A source device (i.e., a smartphone or a PC) cannot output audio to multiple devices simultaneously. So users SHOULD select the output device to use among the available audio output devices
- *Note that this is nothing to do with the 5K input priority. Instead, you must select the audio output device to use through the iOS/Android control panel.



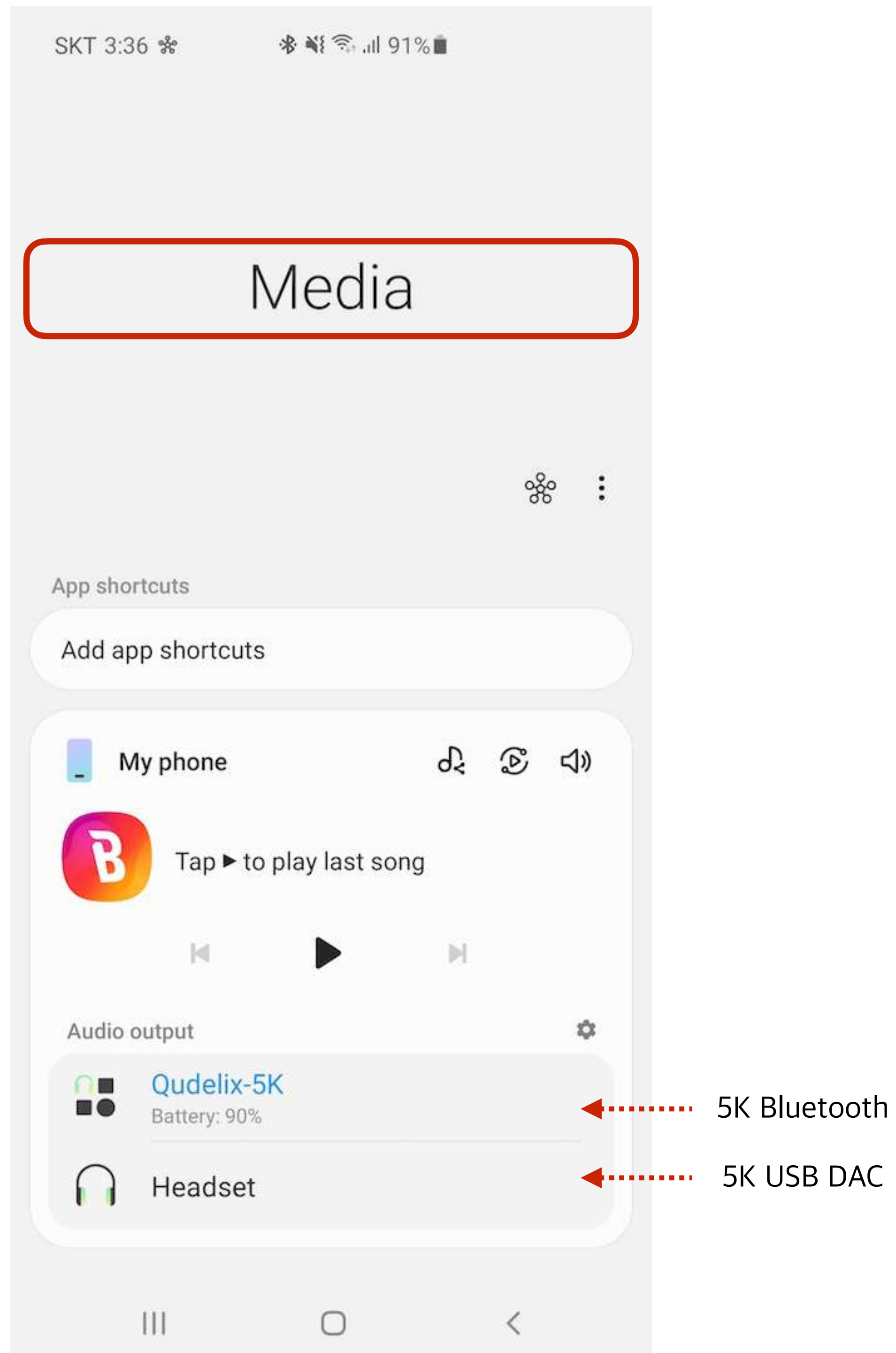
iOS Audio Output Device

- You can simultaneously connect a 5K to an iPhone through Bluetooth and USB (Apple Camera Kit).
- In that case, the iPhone will have multiple audio output devices. And you must select one to use through the iOS control panel.



Android Audio Output Device

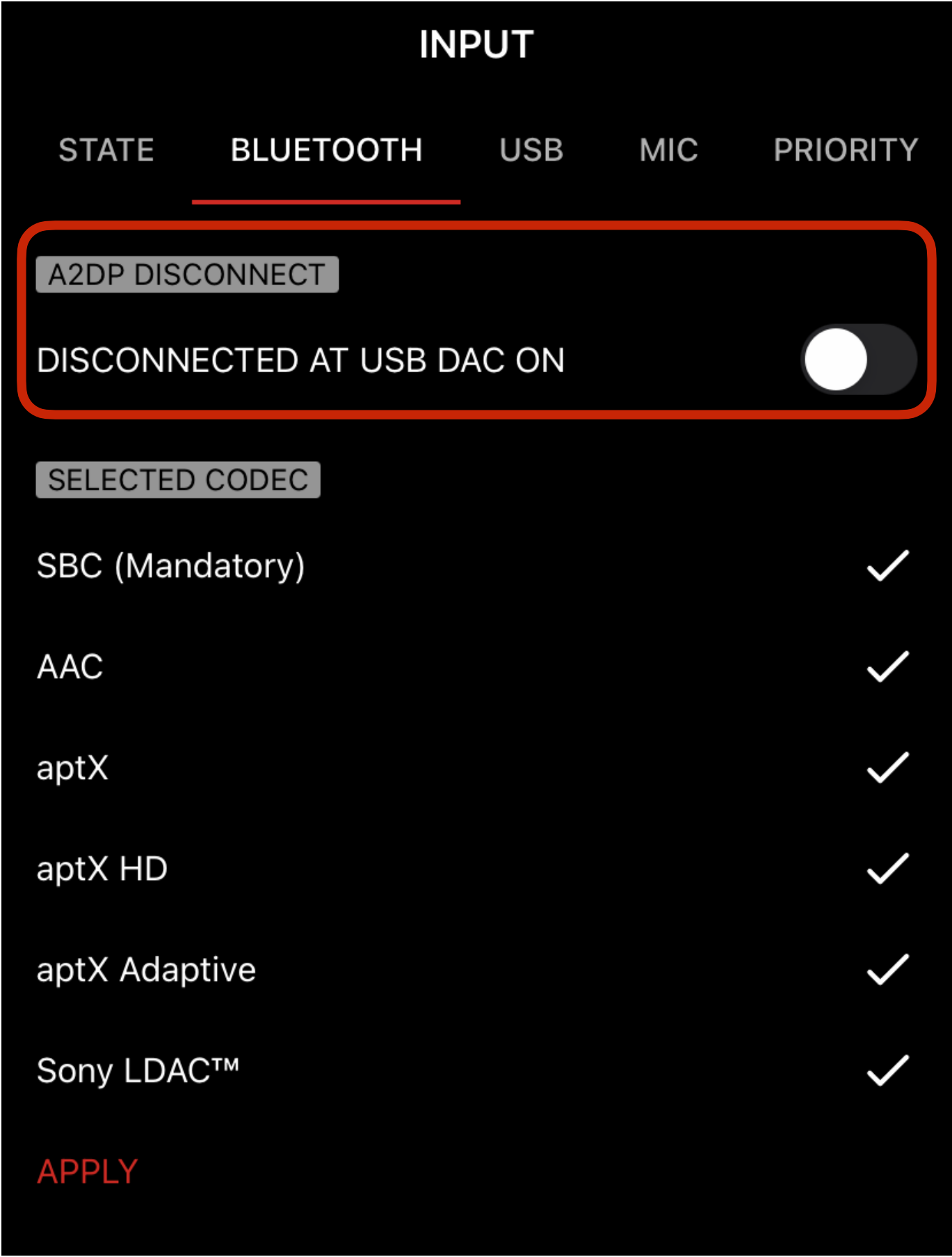
- In Android, Media → Audio Output, you can select an audio output device to use among multiple devices.
- *The Audio Output Selection system menu may differ across smartphone manufacturers, Android versions, and product models.



Android Audio Output Galaxy S10

Bluetooth Auto Disconnect for USB DAC

- 'Bluetooth Auto Disconnect for USB DAC' is a new feature with the software v2.0.
 - 5K automatically disconnects any existing Bluetooth Audio Link when connected to a USB Host. And if the USB is disconnected, 5K reconnects the Bluetooth Audio Link.
 - It only disconnects the Audio Link so that you can keep the Qudelix iOS/Android App and the voice call over Bluetooth.
- For example,
 - When you plug a 5K into a smartphone over USB, the 5K automatically disconnects the Bluetooth Audio link.
 - Then, you unplug the 5K from the smartphone, and the 5K automatically reconnects the Bluetooth audio link.
 - This way, the smartphone will have only one audio output device at a time, the 5K Bluetooth or the 5K USB. It can save you from the tedious work of selecting an audio output device repeatedly.

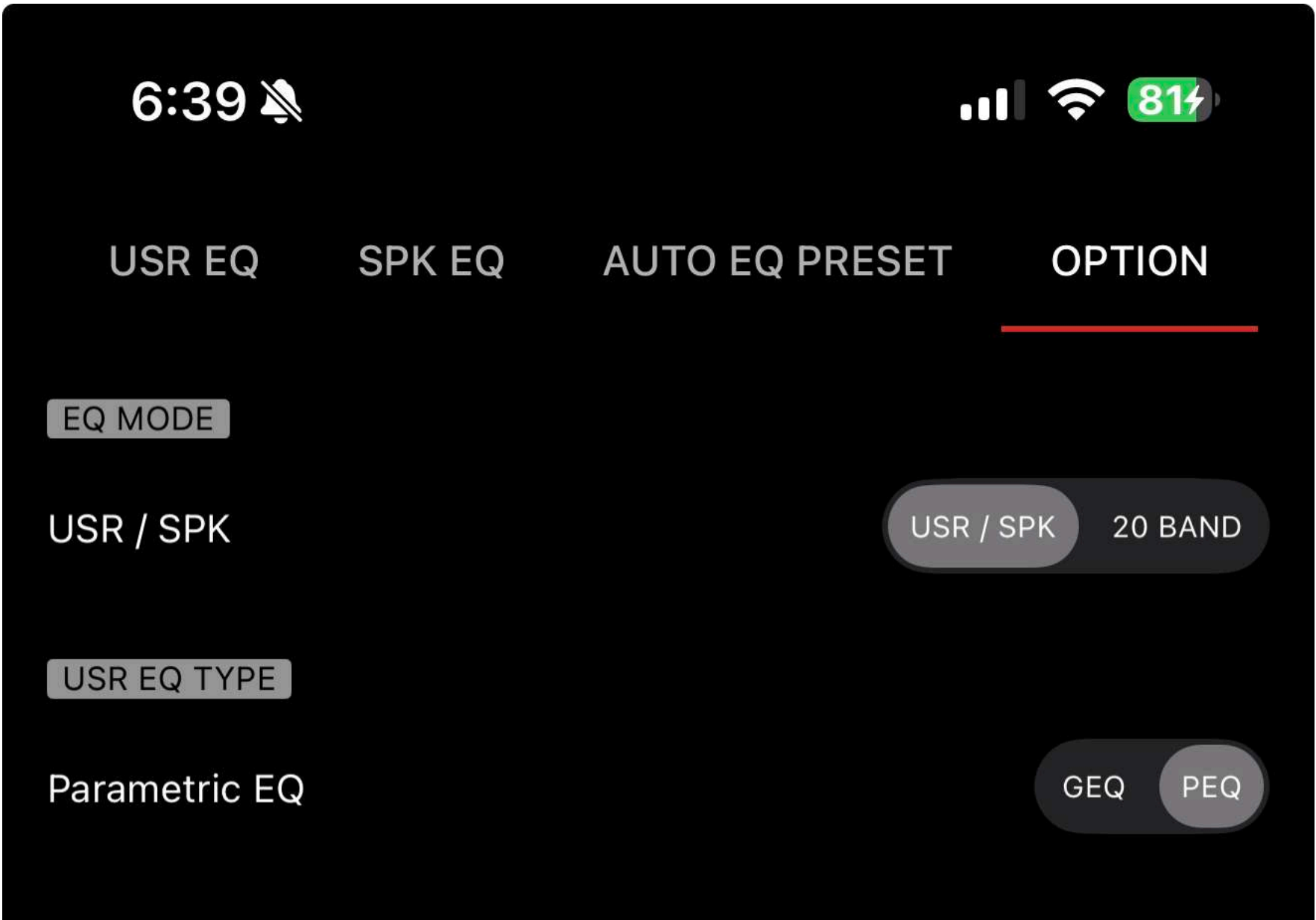


Equalizer

- The embedded pro-grade equalizer is a vital benefit of 5K DAC/AMP over other similar products. It processes every incoming audio from everywhere: Youtube, Apple Music, Spotify, iOS, Android, PC, Bluetooth (all Codecs), and USB.
- 5K DAC/AMP supports parametric EQ and graphical EQ.
- The embedded DSP cores perform double-precision 64-bit EQ processing, providing the most accurate and precise frequency response even at low-frequency range.
- 5K manages and stores the EQ presets in its internal flash memory. And 5K automatically reloads the settings at boot-up and resumes the same EQ setting as configured before. So, once you configure the EQ using the app, you no longer need to launch the app to see if the EQ runs or not.
- EQ modes
 - 10-Band USR/SPK Equalizer
 - 20-Band Equalizer

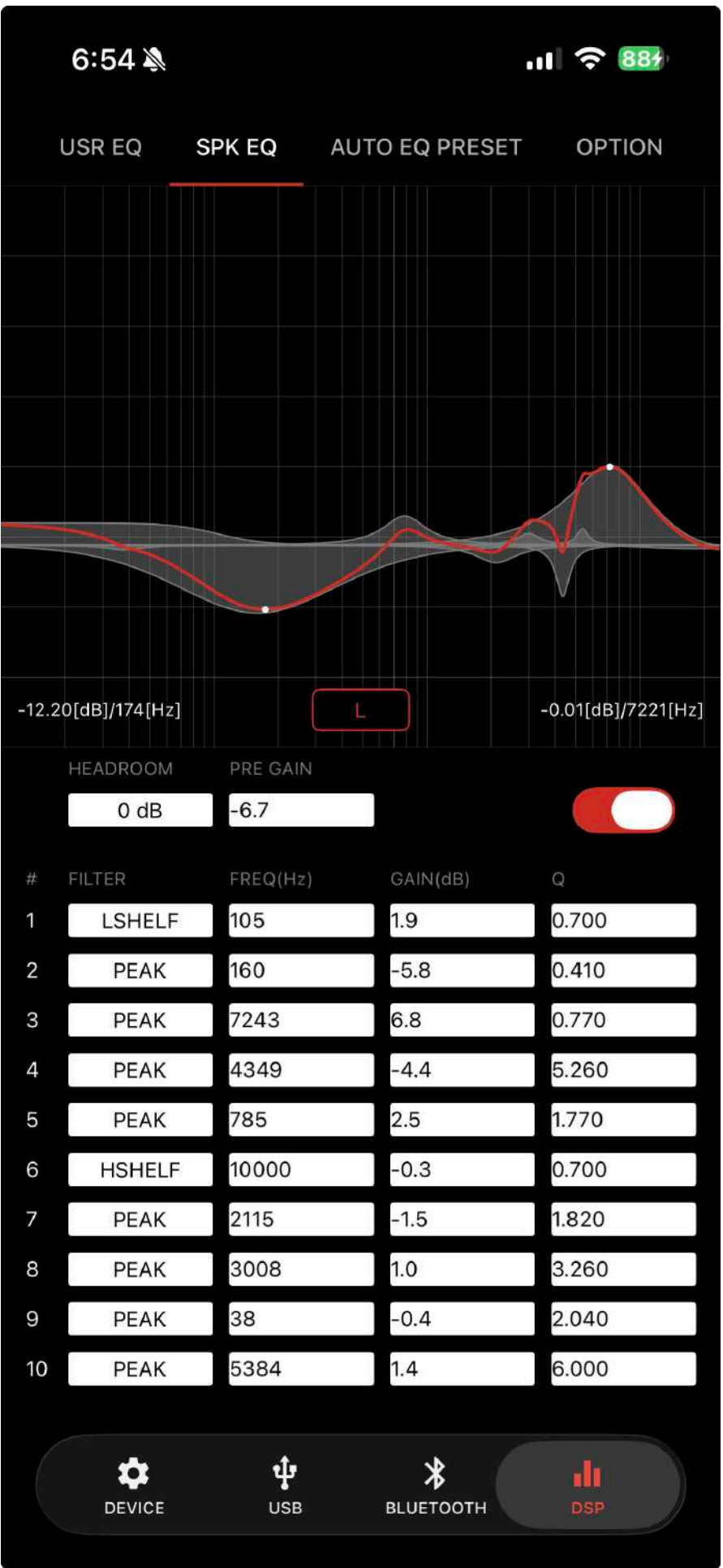
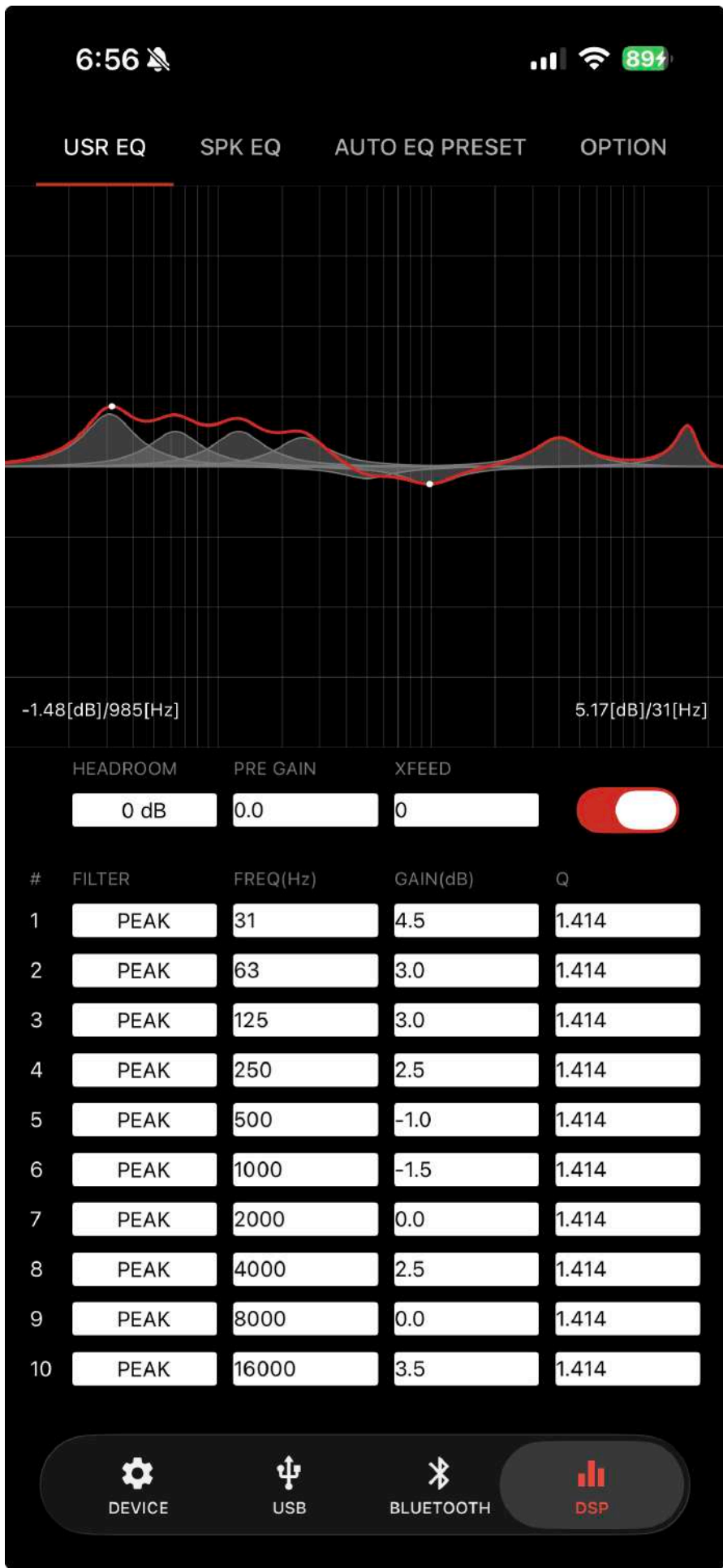
Equalizer Mode : USR/SPK

- You can select the EQ mode in the Qudelix app under:
App → DSP → OPTION → EQ MODE
- USR / SPKEQ MODE provides two independent logical 10-band EQs:
 - **USR EQ** (User EQ)
 - **SPK EQ** (Speaker/Headphone EQ)
- These two EQs are cascaded and processed sequentially.
- Although they are physically implemented using the same DSP EQ engine, they are logically separated according to their intended purposes.
- **SPK EQ** is recommended for compensating or tuning headphones, earphones, or speakers.
 - Supports independent L/R channel EQ adjustment
 - AutoEq presets are applied to the SPK EQ section
- **USR EQ** is recommended for adjusting tonal preference according to:
 - Music genre
 - Audio source characteristics
 - Personal listening preference
- This layered EQ structure allows independent device correction and user sound customization simultaneously.



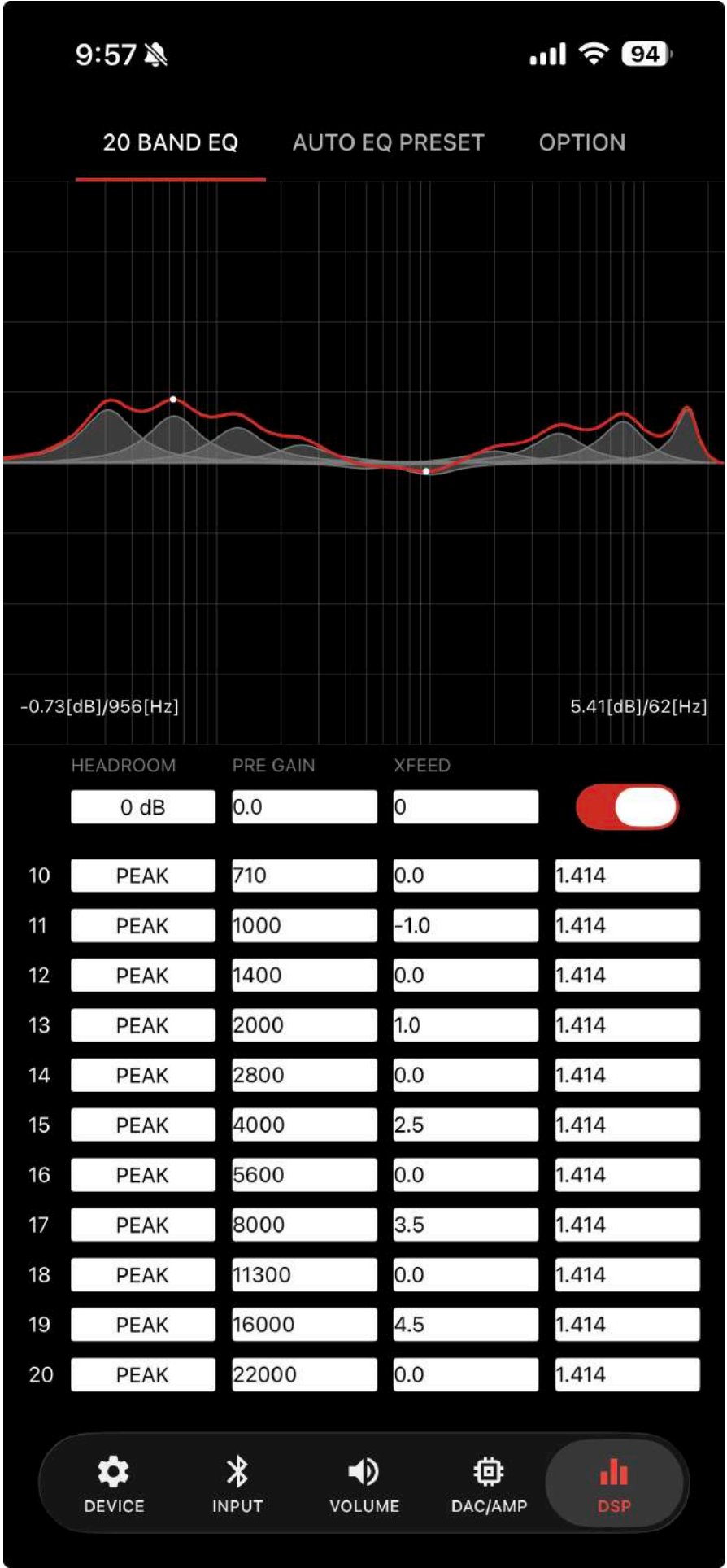
Equalizer Mode : USR/SPK

- **Crossfeed (XFEED)**
 - The USR EQ page also provides an adjustable XFEED (Crossfeed) control.
 - Crossfeed blends a portion of the left and right audio channels to simulate a more natural speaker-like listening experience when using headphones.
 - 0 = XFEED Off
 - 50 = Full XFEED (Mono)
 - Higher XFEED levels increase channel blending between L/R channels.
 - Crossfeed may improve listening comfort during long headphone listening sessions.
- **Independent L/R EQ Adjustment**
 - SPK EQ supports independent left/right (L/R) channel EQ adjustment.
 - This feature can be used to:
 - Compensate for headphone or earphone channel imbalance
 - Compensate for differences in left/right hearing sensitivity
 - Independent L/R adjustment allows more precise channel matching and personalized listening correction.



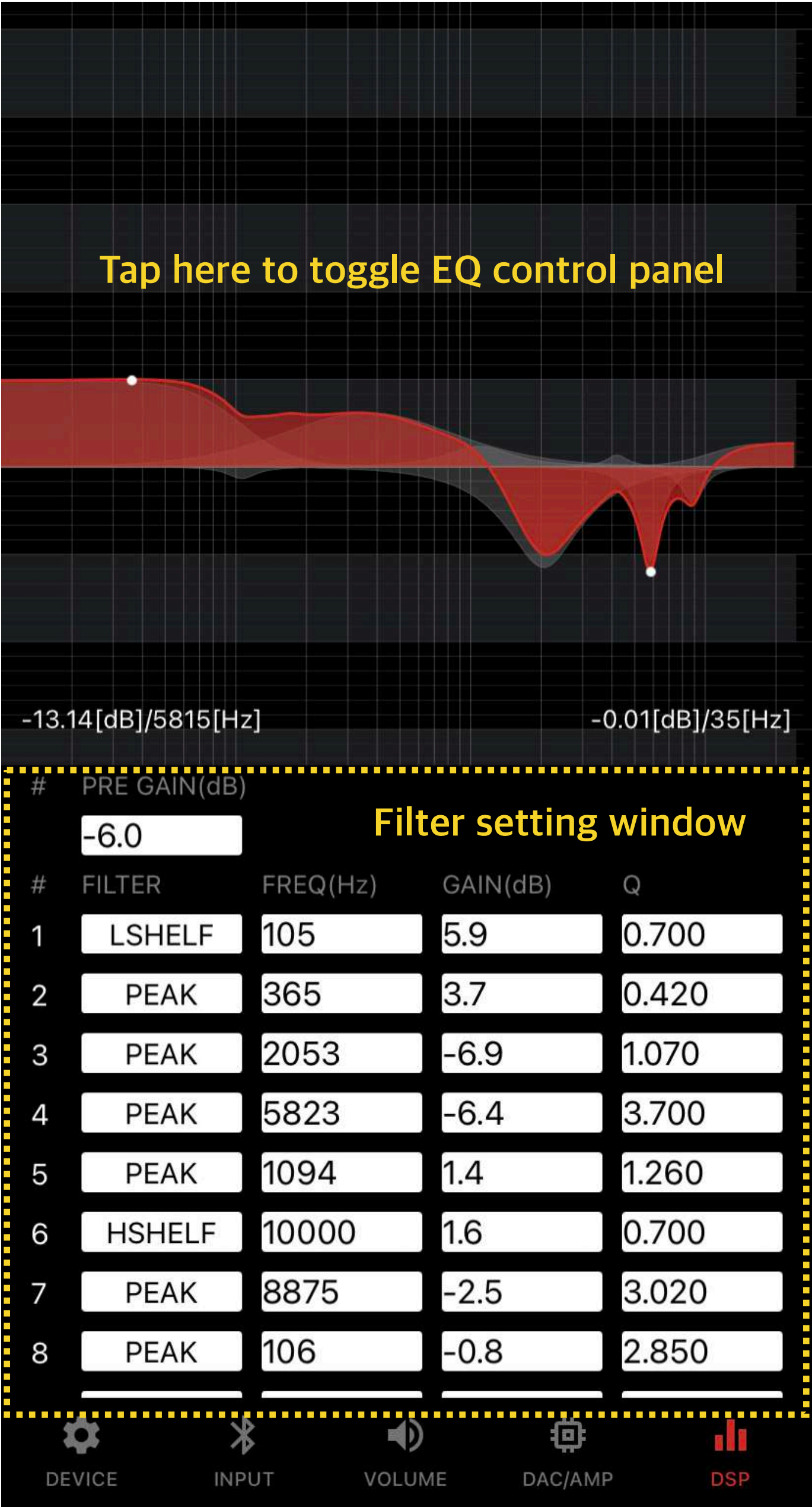
Equalizer Mode : 20-Band

- **20-Band EQ Mode**
 - When 20-Band EQ mode is selected, a single 20-band EQ stage is available.
- Both the USR/SPK EQ mode and the 20-Band EQ mode physically provide 20 EQ filters in the DSP engine.
 - However, their management structures are different.
- USR/SPK EQ mode
 - Organizes the 20 filters into two logical 10-band EQ layers
 - Separately stores and manages:
 - USR EQ (User EQ)
 - SPK EQ (Speaker/Headphone EQ)
 - SPK EQ (Speaker/Headphone EQ) allows independent L/R EQ adjustment and separate EQ preset management for each channel.
- 20-Band EQ mode
 - Uses all 20 filters as a single EQ stage
 - Manages the entire 20-band configuration as one EQ preset



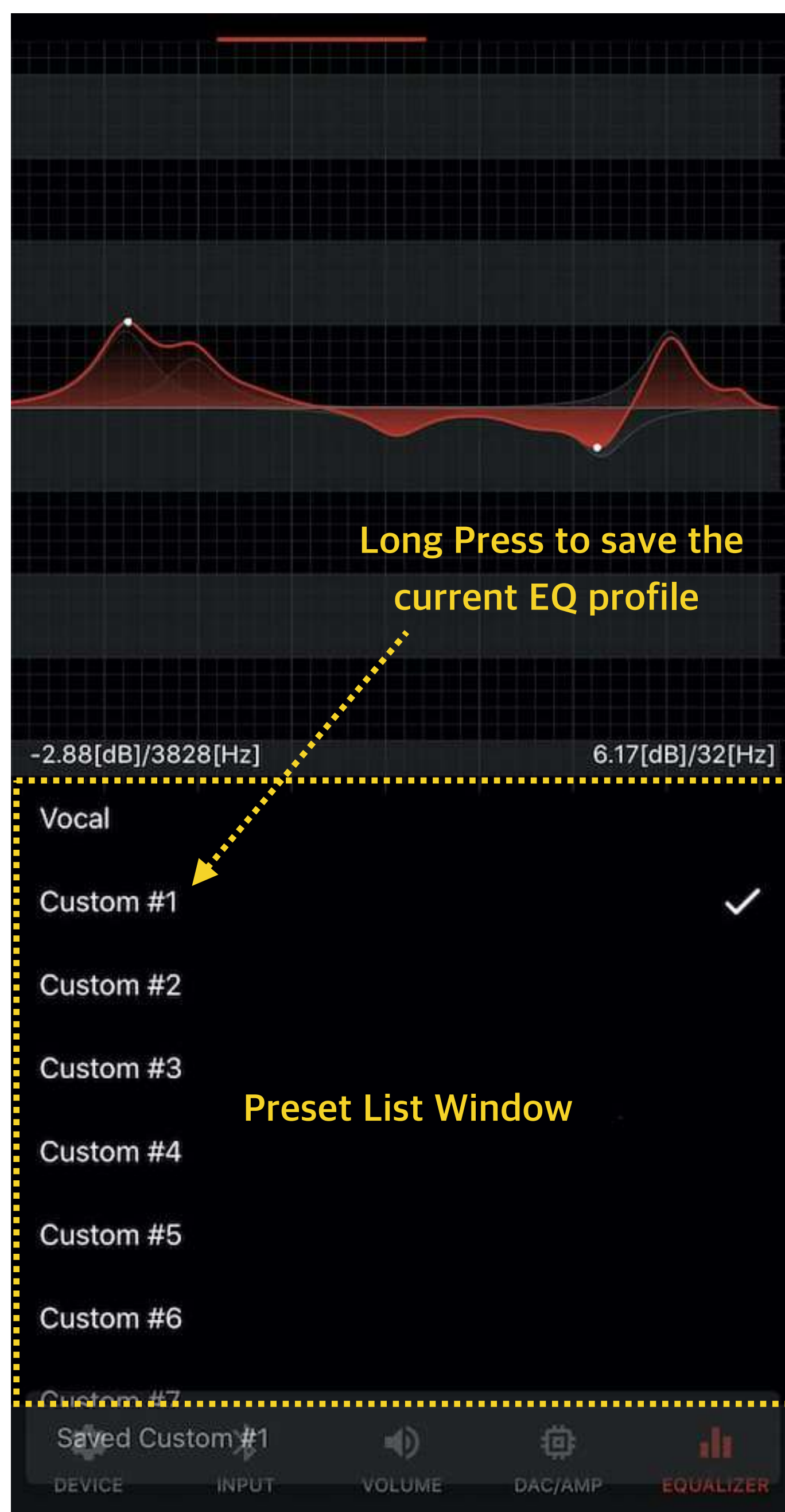
Filter Setting Window

- Tapping the EQ Graph pane will toggle the EQ control panel.
- Filter setting window
- Preset list window



Preset List Window

- Up to 20 Custom User Presets
- 5K manages and stores the EQ presets in its internal flash memory.
- **Long-press (iOS/Android App) or Double-click (Chrome App) custom#1~#20 will show you the popup window saving the preset.**

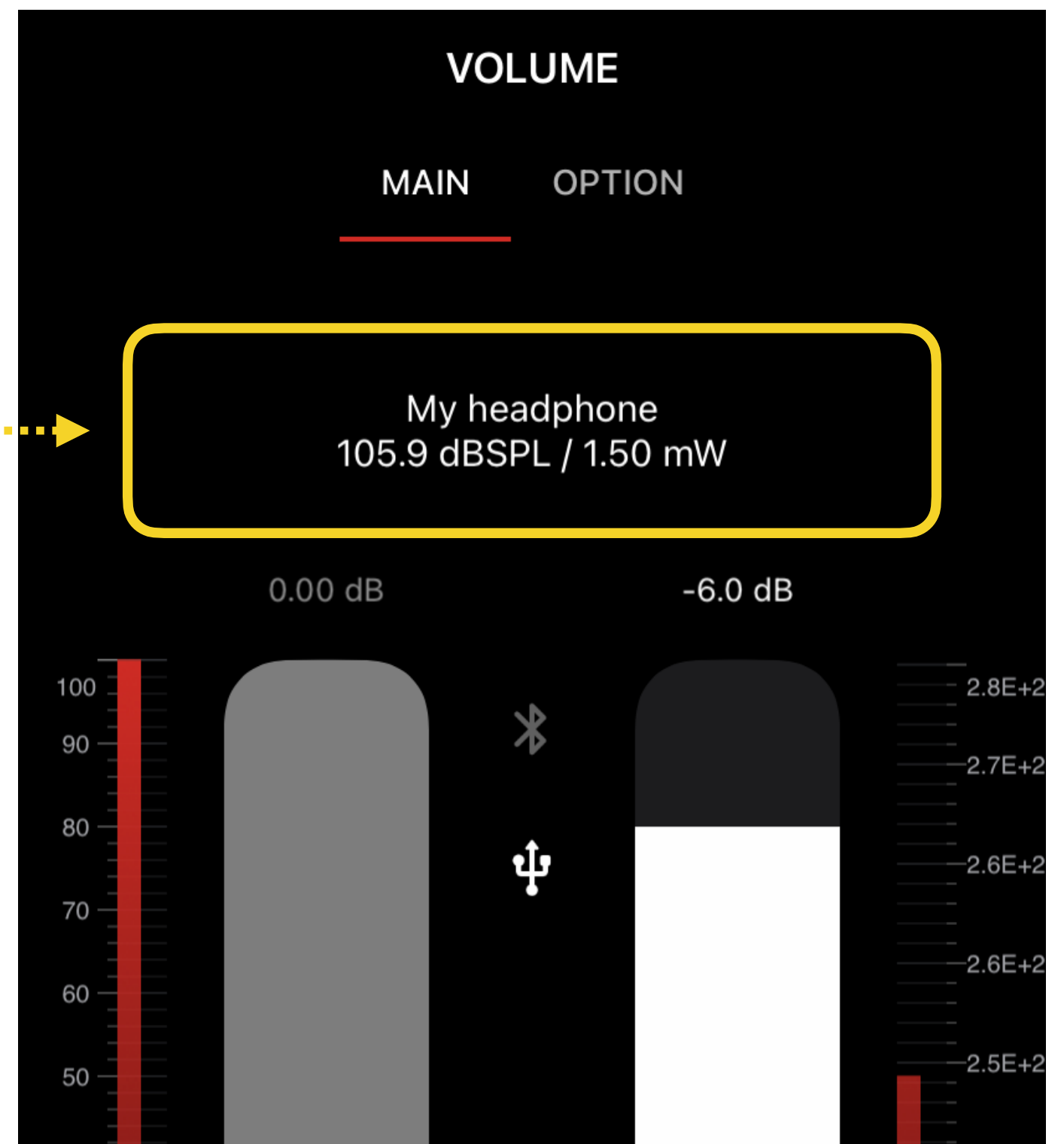
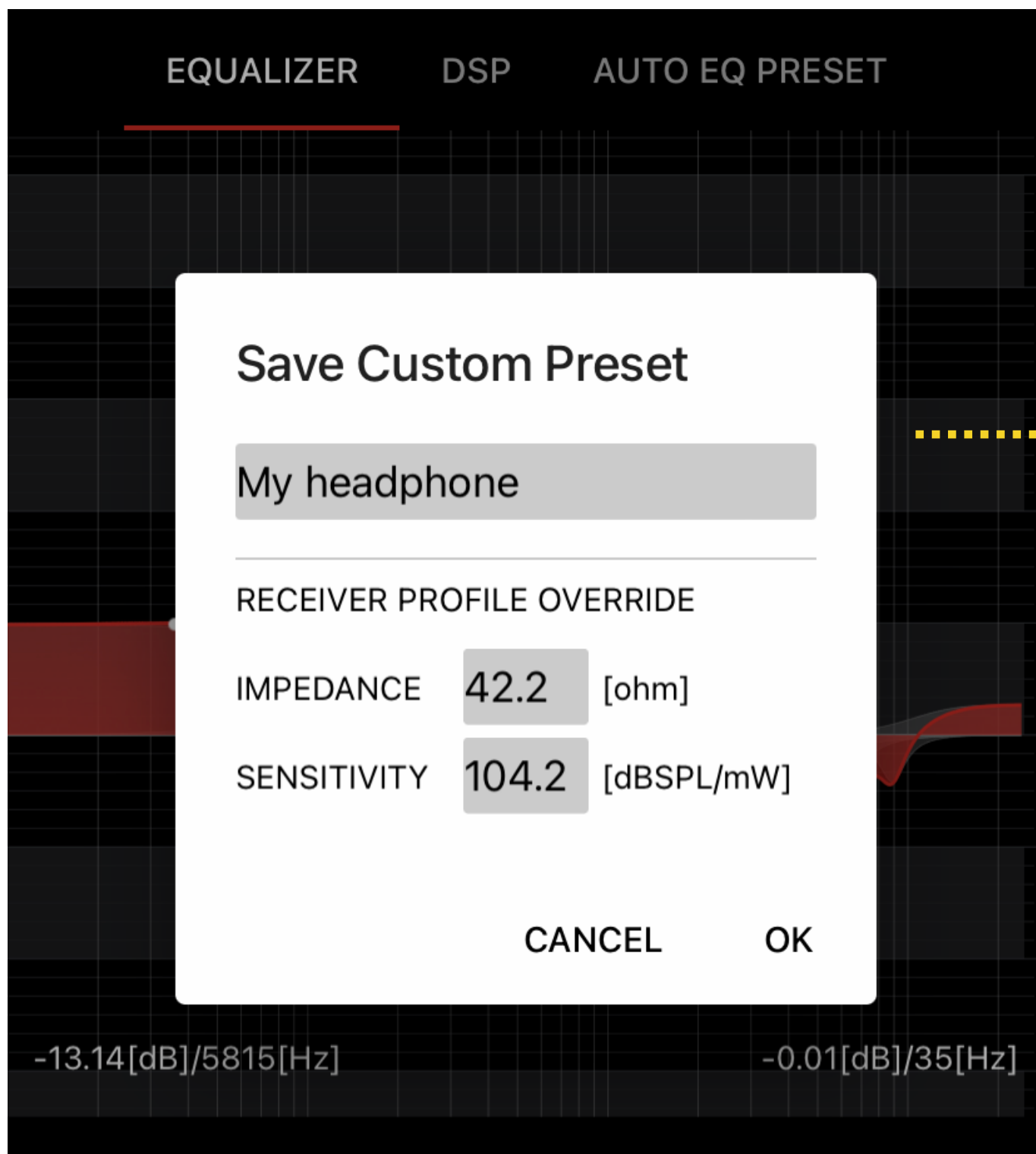


EQ Headroom & Pre Gain

- When an EQ filter gain is higher than 0 dB, the filter boosts up the input audio, and the audio may get clipped/saturated, resulting in unwanted clipping noise.
- So, we must scale down the input audio before EQ filters, making the frequency peak of an EQ preset no higher than 0dB.
- The Preamp Gain and Headroom are the scale factors applied to the input before EQ filtering.
 - **Scale factor [dB] = Preamp Gain [dB] + Headroom [dB]**
- The EQ headroom is a global factor applied to all EQ presets. And the EQ frequency plot, min/max display, does not reflect the headroom. In other words, the headroom would change the audio output level audibly but not the frequency plot visually.
- Meanwhile, the Preamp Gain works for each EQ preset. And it visually changes the EQ frequency plot and audibly changes the audio output level.

Saving EQ preset

- Long-press (iOS/Android App) or Double-click (Chrome App) custom#1~#20 will show you the popup window for saving the preset.
- When you save an EQ preset, you can save the impedance and sensitivity profile along with the EQ preset.
- Technically, EQ and Impedance/Sensitivity have nothing to do with each other. However, since most users manage their EQ preset per IEM/Headphone basis, we added the impedance/sensitivity fields in each EQ preset.
- We highly recommend setting Impedance/Sensitivity, but you can leave them as zeros.



- AutoEQ is an open-source project.
 - <https://github.com/jaakkopasanen/AutoEq>
- AutoEQ analyzes the characteristics of earphones and headphones through actual measurements using specialized equipment. Based on this data, it creates EQ presets that bring the sound closer to a reference target. The Qudelix app integrates this vast database of AutoEQ presets, providing a user interface that allows these presets to be applied directly to the device with simple adjustments.
- From an analog and acoustic perspective, it is nearly impossible to achieve perfect response across all frequency ranges for all earphones and headphones. Audio engineering always involves performance trade-offs. Using AutoEQ to compensate for shortcomings in well-designed earphones and headphones is a very reasonable approach. Especially for digital audio sources, a precise DSP EQ processor has virtually no audible degradation on sound quality.
- *Some may think that AutoEQ adjusts the sound to match the earphones or headphones they are using, but this is a misunderstanding. Only AutoEQ presets that specifically match the exact model of your earphones or headphones should be selected/applied.
- In the AutoEq page, selecting a headphone or earphone preset automatically applies the EQ settings to either:
 - 20-Band EQ, or
 - SPK EQ
 - depending on the currently selected EQ mode.
 - This allows AutoEq compensation profiles to integrate directly with the active EQ processing structure.
 - Qudelix recommends searching for your headphone or earphone model in the AutoEq database, applying the matching AutoEq preset, and saving the applied EQ preset for convenient future use.

Mic. Pass-Thru Mode

- MIC PASS-THRU mode
 - Adjust Mic. Gain
 - Adjust the mix-up ratio with AMBIENT LEVEL slider
- If ABSOLUTE VOLUME is ON,
 - The smartphone side-key volume adjusts the overall 5K device volume.
 - The smartphone side-key volume adjusts both microphone and music at the current mix-up ratio.

5:21

INPUT

STATECODECUSBMIC

MICROPHONE GAIN 22

01122

MIC SELECTION

5K Built-In MEMS Mic

Earphone Inline Mic

MIC PASS-THRU

Voice Call

Music

AMBIENT LEVEL 0

-50050

Microphone Pass-Through mode enables you to hear the ambient sound when in the music streaming and voice call. During the phone call, it helps you to listen to your voice. AMBIENT LEVEL adjust the mixup ratio between Microphone and Music. *For Voice Call, the ambient mixup ratio is fixed to 50:50.

DEVICE

INPUT

VOLUME

DAC/AMP

EQUALIZER

Mic. Only

Mic. GAIN

Music Only

Antenna

High Sensitivity 3-dimensional LDS Antenna is printed on the bottom case and is placed around the red LED button. 5K provides the best RF performance in most use cases. However, for the optimal performance, please keep this area away from any metallic objects.

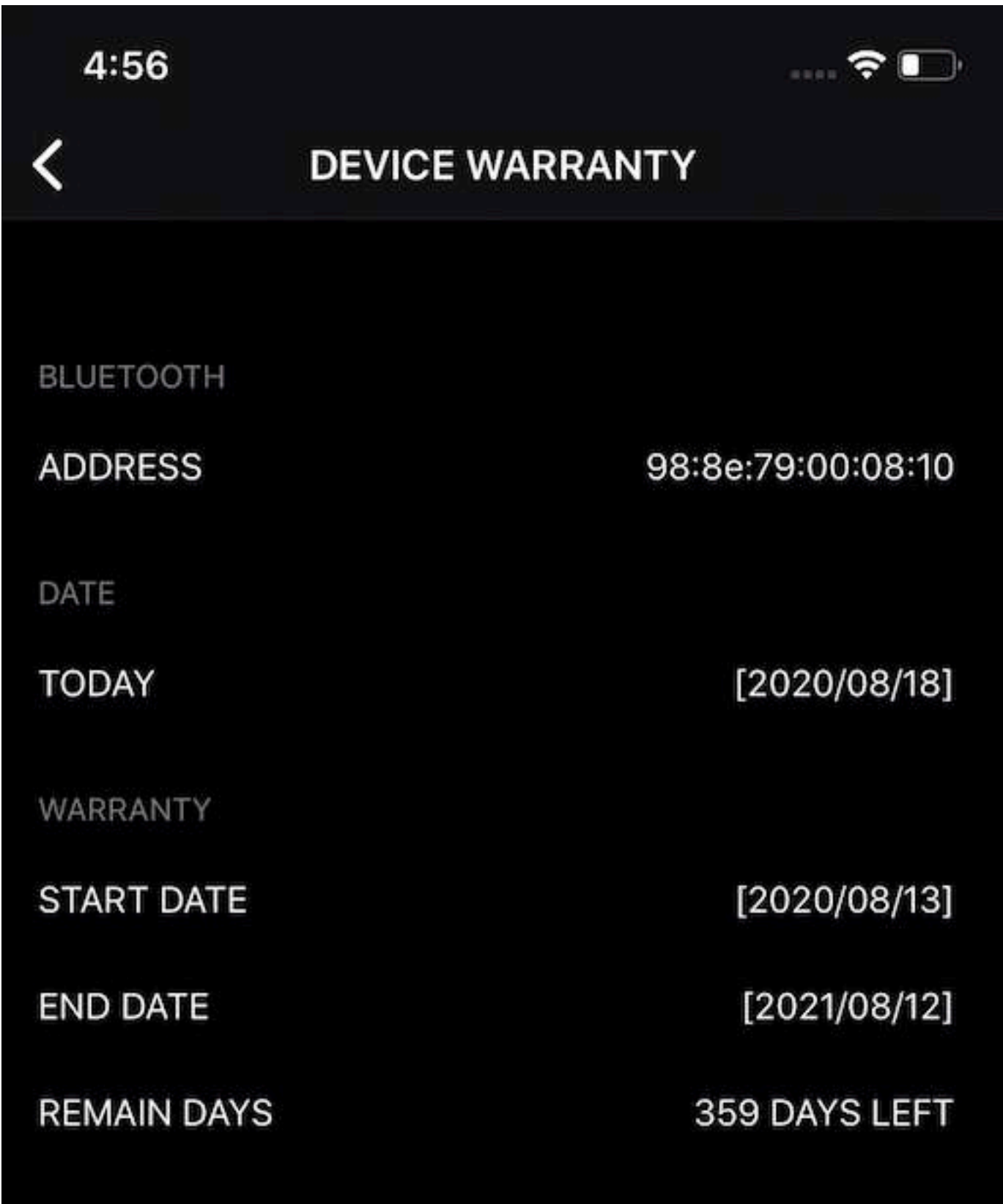
Bluetooth 2.4 GHz Class 2.5mW (4dBm)

3 Dimensional LDS (Laser Direct Structuring) Antenna



Warranty

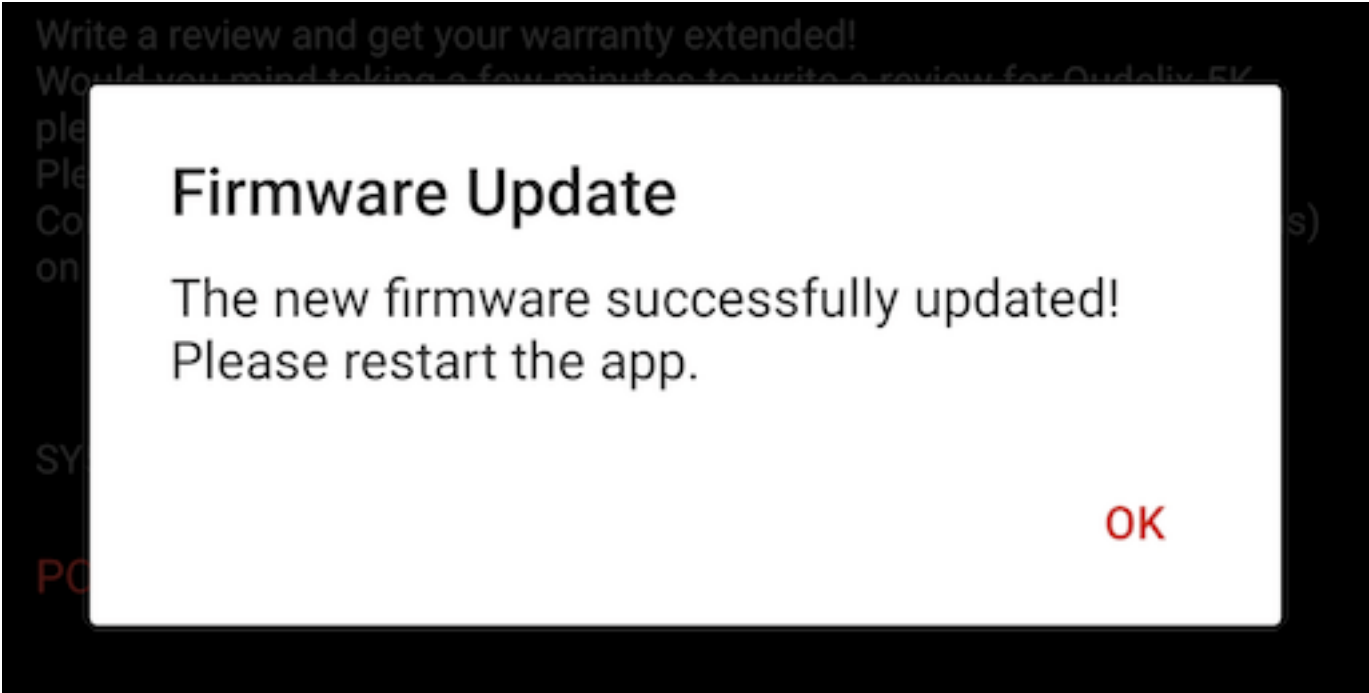
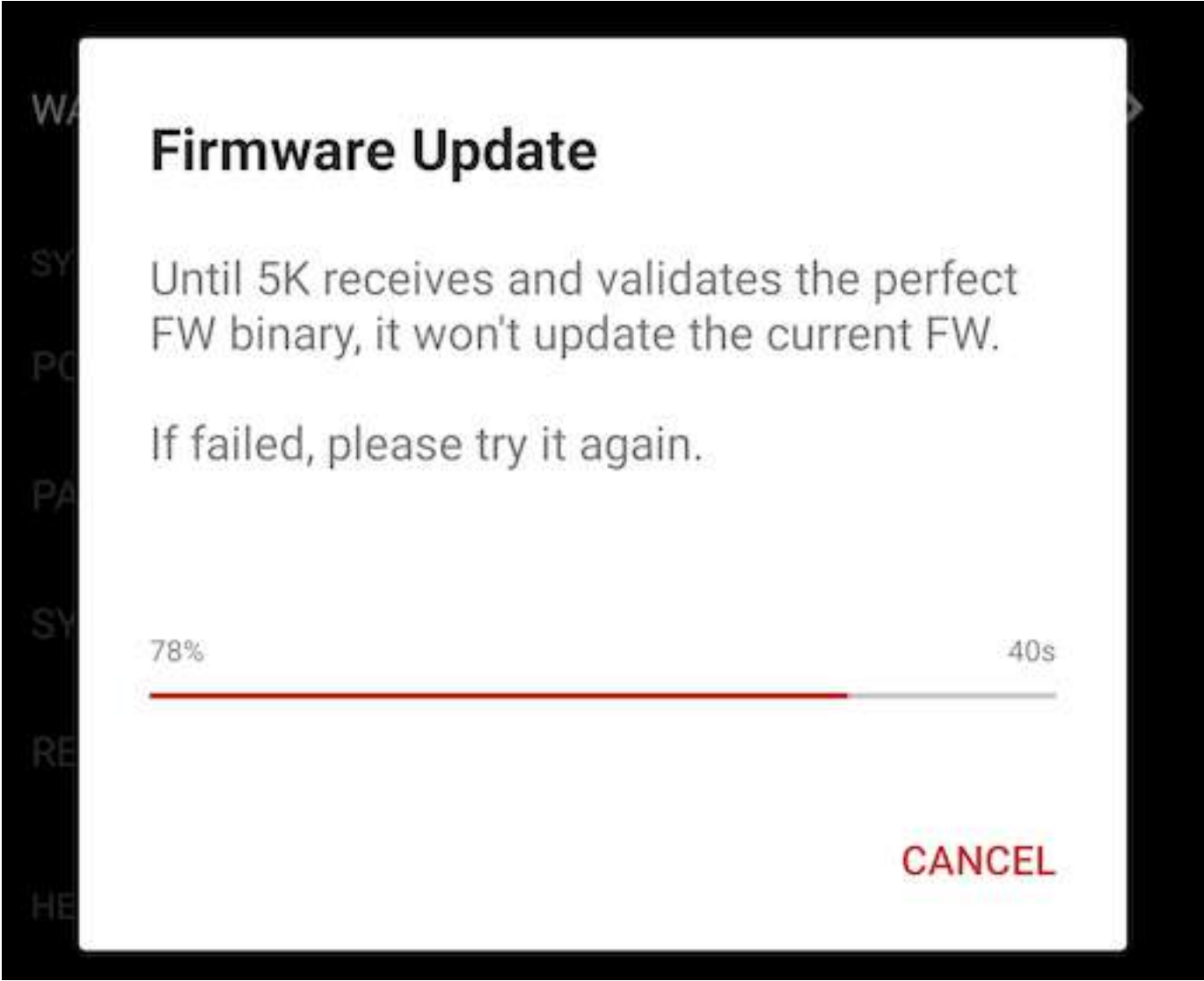
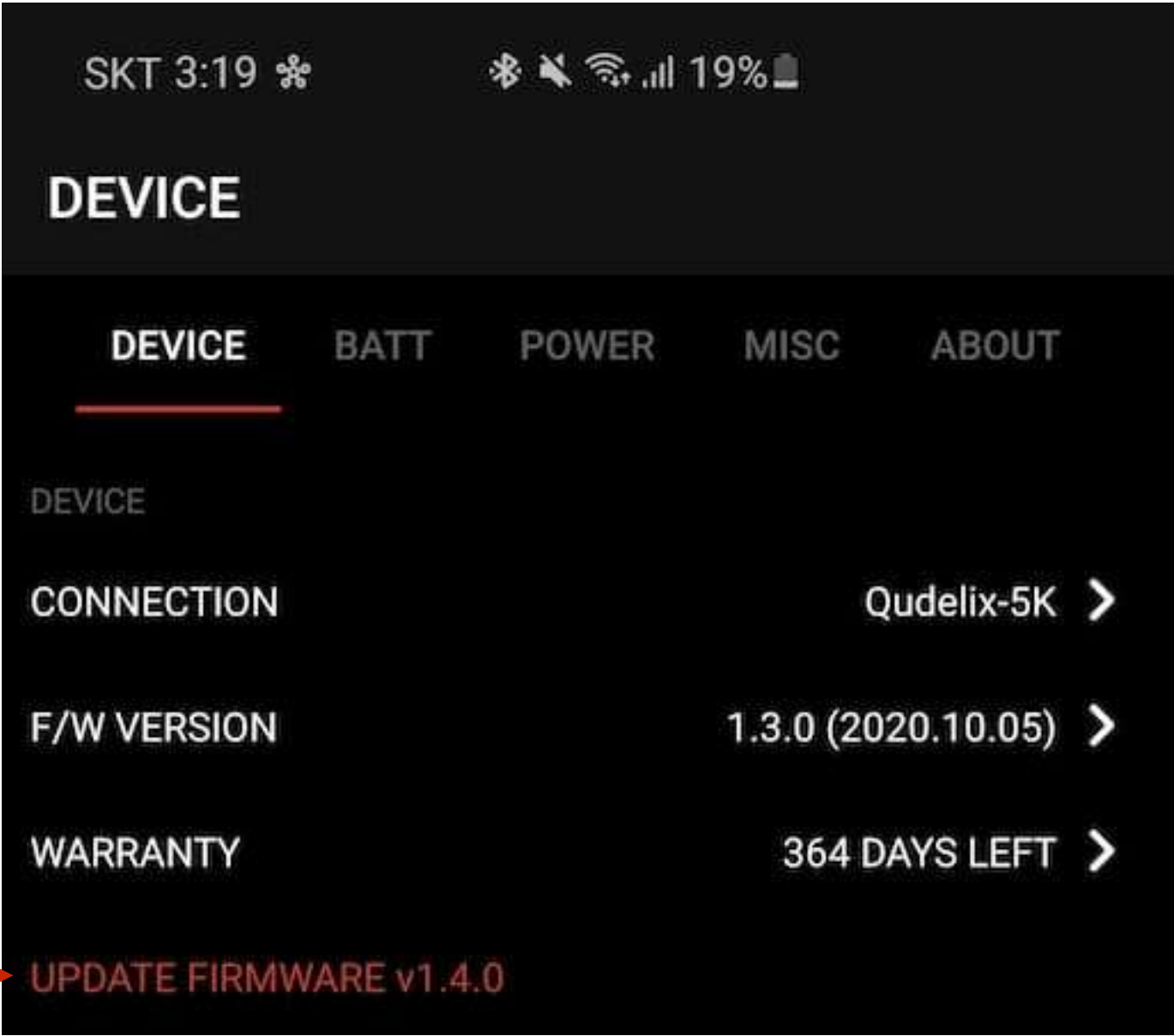
- **One year warranty**
 - From the date of the first app connection
 - Automatically activated from the first connection with the Qudelix App
 - Any problem caused due to a defect in manufacturing will be repaired or replaced
 - The warranty is transferable and is not limited to the original purchaser
- **Warranty is managed upon Bluetooth MAC address and serial number**
 - No need to keep the proof of the purchase
 - No receipt or invoice is required.
- **Any issue caused by accident or misuse**
 - Can be replaced with a refurbished product at customer’s charge & shipping fee
- **For any purchase from local distributors**
 - The respective distributor is responsible for the warranty and will handle the case directly
- **WARNING**
 - Warranty void if the case opened
 - DO NOT disassemble or open the case.



Firmware Update - OTA

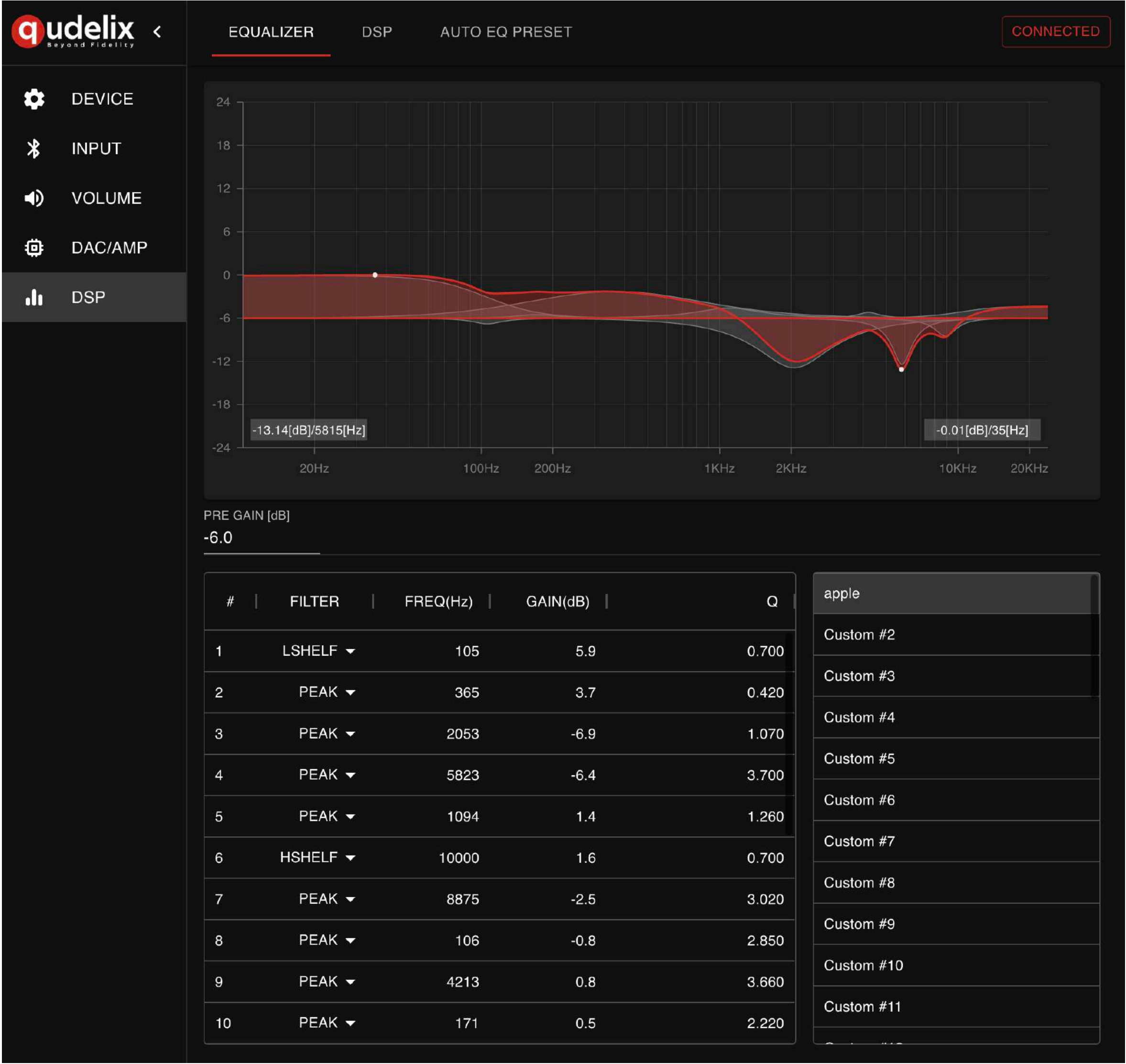
- **OTA(Over The Air) Firmware Update**
 - The app updates the 5K firmware over the Bluetooth link.
 - **UPDATE FIRMWARE vX.X.X** will appear when a new FW is available.
 - Please keep the 5K close to the smartphone during the FW update.
 - Until the 5K fully receives and validates the firmware binary, it won't update the current FW.
 - If failed, please try it again.

FW Update



Qudelix Chrome Extension

- <https://chrome.google.com/webstore/detail/qudelix/phaodiidhofhdmfkjiacigibgikhfafn>
 - For a USB DAC use case with Windows, macOS, and Linux, you can install the Qudelix Chrome Extension.
 - And you can simultaneously keep the app connections with a smartphone over Bluetooth and a PC over USB.
- *The Chrome app requires the 5K firmware version 1.12.0 or higher, please update the 5K device firmware first using the mobile app.



Appendix - PCBA

5K has three BGA(Ball Grid Array) IC parts; QCC5124, and two ES9219

To enhance PCB reliability and durability, epoxy underfill was applied on those sensitive BGA parts, despite the additional manufacturing cost. Those components are mechanically reinforced by both solder joints and epoxy underfill.



Appendix - Size & Weight

Size

52.8(H) x 26.7(W) x 15.6(D) mm (including Clip)

Weight

25 g



Appendix - Package

PACKAGE

91mm (L) x 55mm (W) x 55mm (H)

INCLUDED

USB C-to-C Cable (120mm)

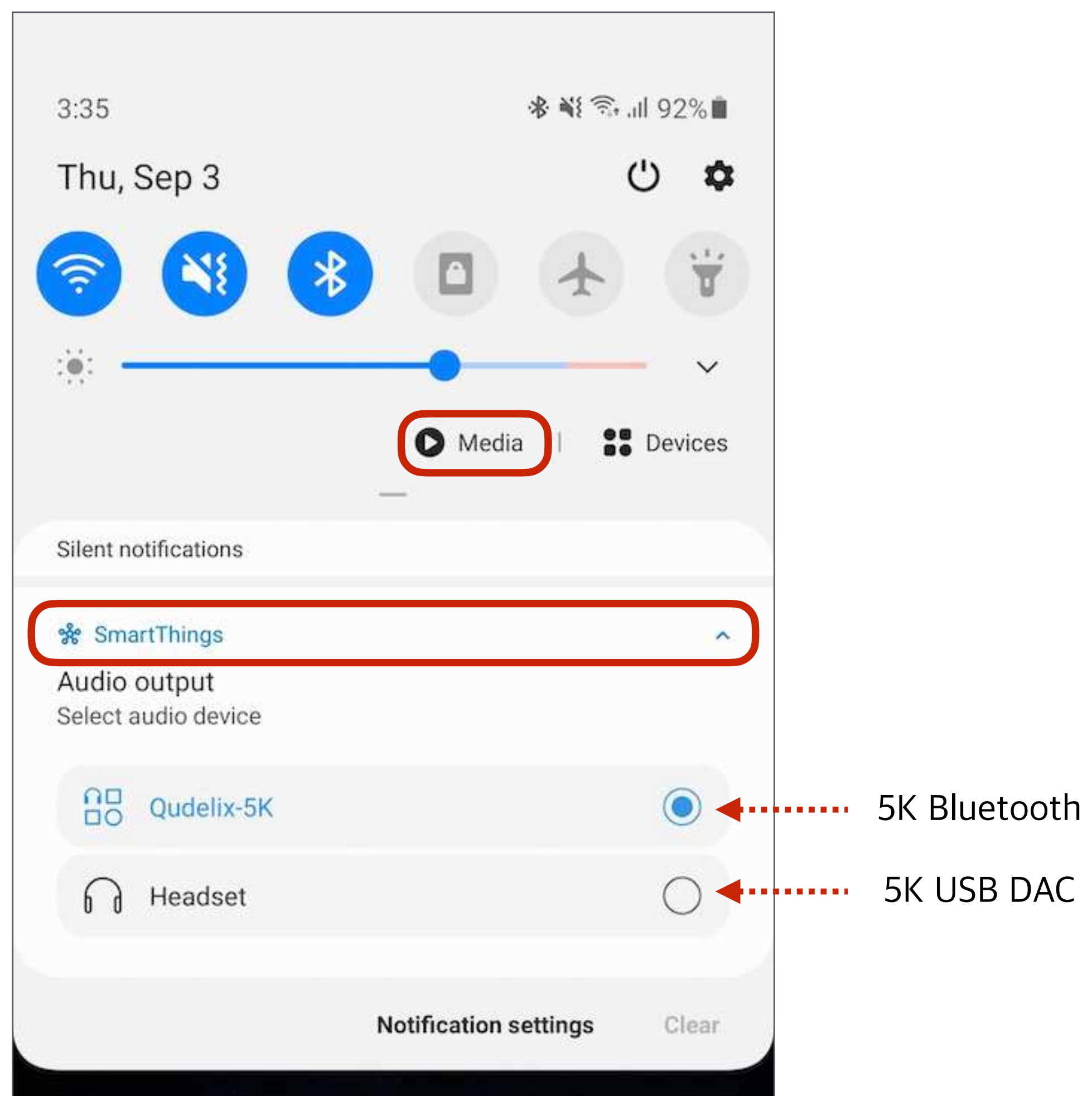
USB C-to-A Cable (120mm)

*No User Manual included in the package. Please refer to this user guide.



Appendix - Android Audio Output Device

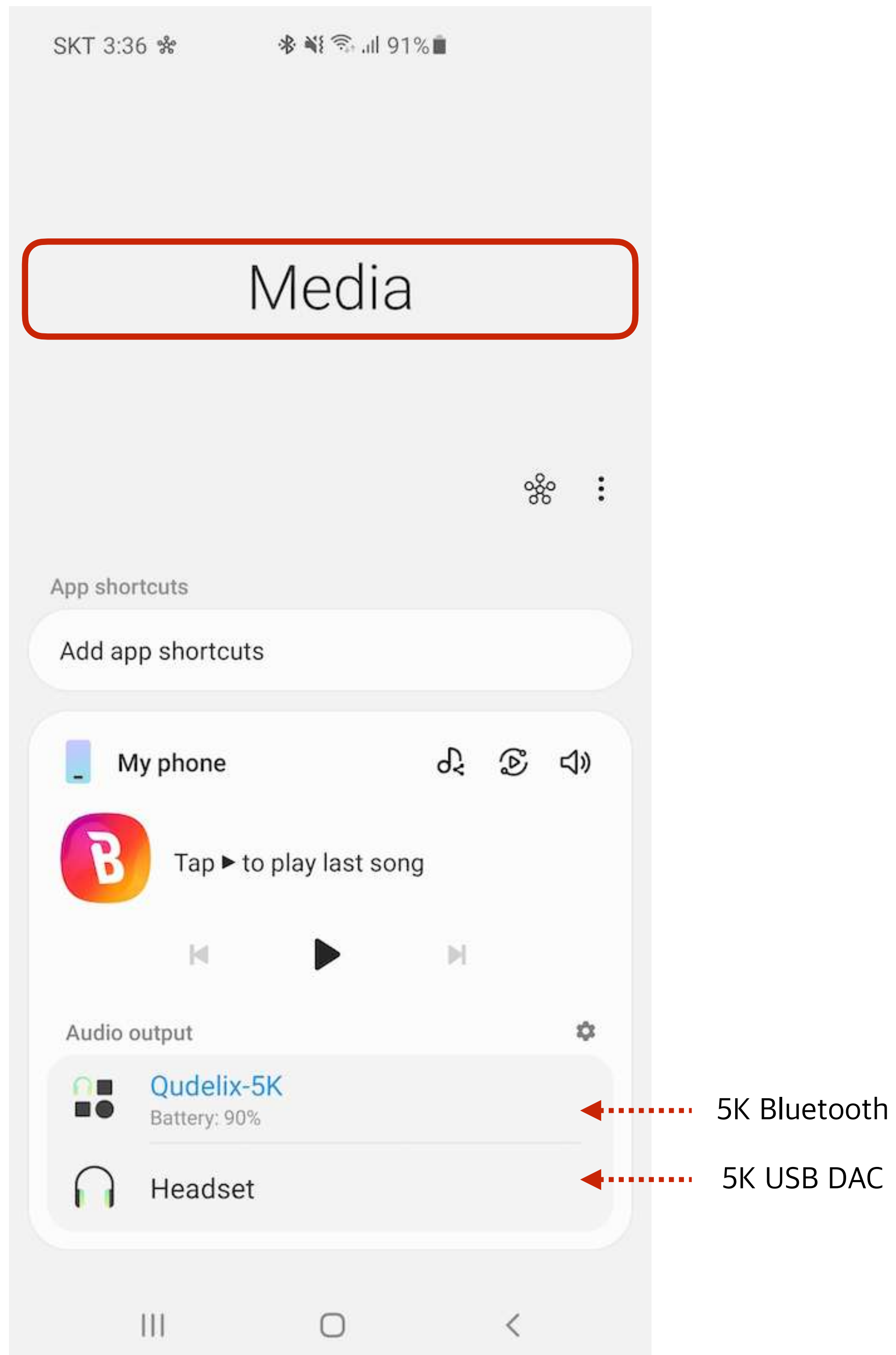
- 5K can be connected to Android both through Bluetooth and USB DAC (C-to-C) at the same time.
- In that case, the Android device detects multiple audio output devices. And the Android selects the active output device exclusively.
- *With some specific Android devices, **Sony audio app** sometimes overrides the system Audio Output selection to Bluetooth forcibly. If you don't get any sound through Android USB DAC, please try uninstalling Sony audio app.



Android Audio Output Galaxy S10

Appendix - Android Audio Output Device

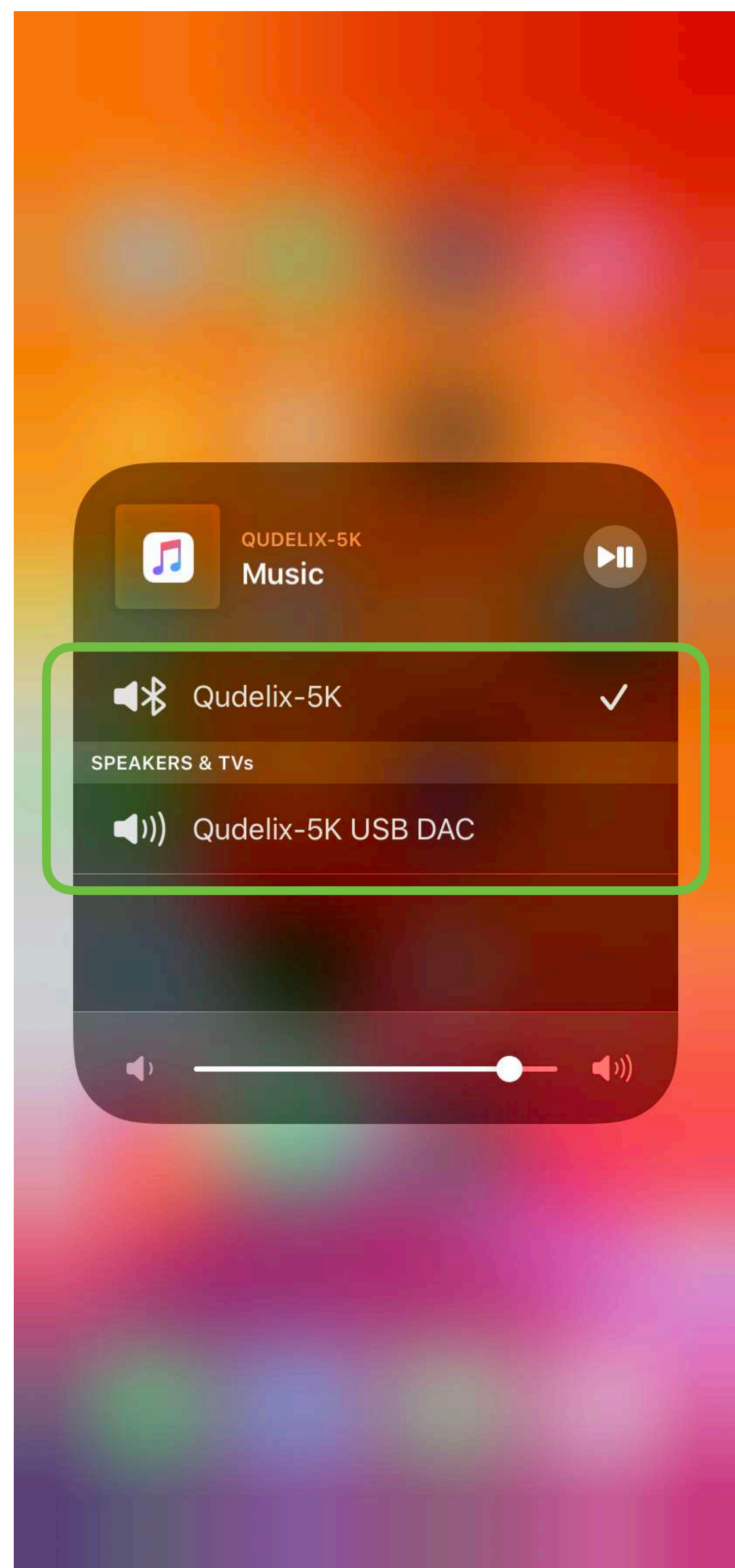
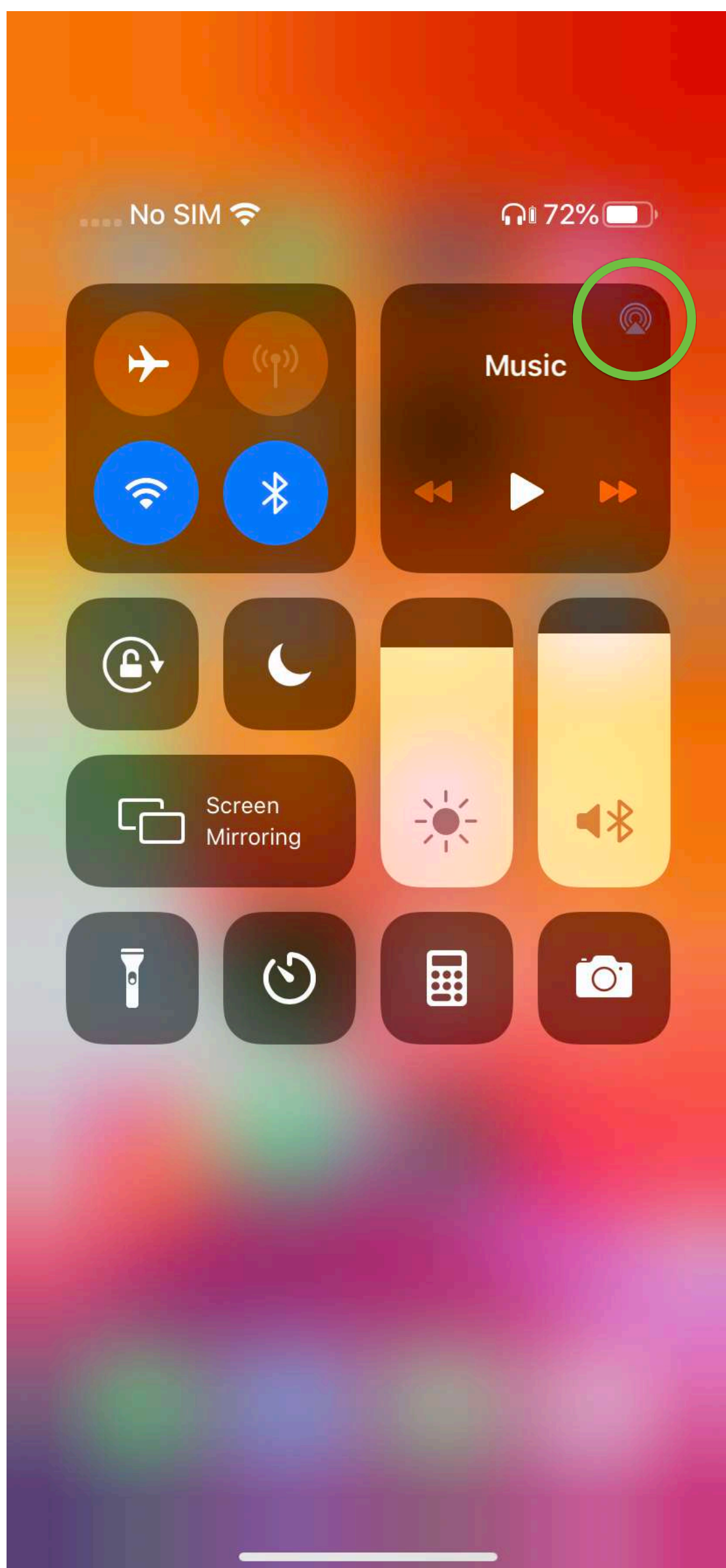
- In Android, Media → Audio Output, you can select the active device you want to use.
- The system menu for Audio Output may be different across smartphone manufacturers.



Android Audio Output Galaxy S10

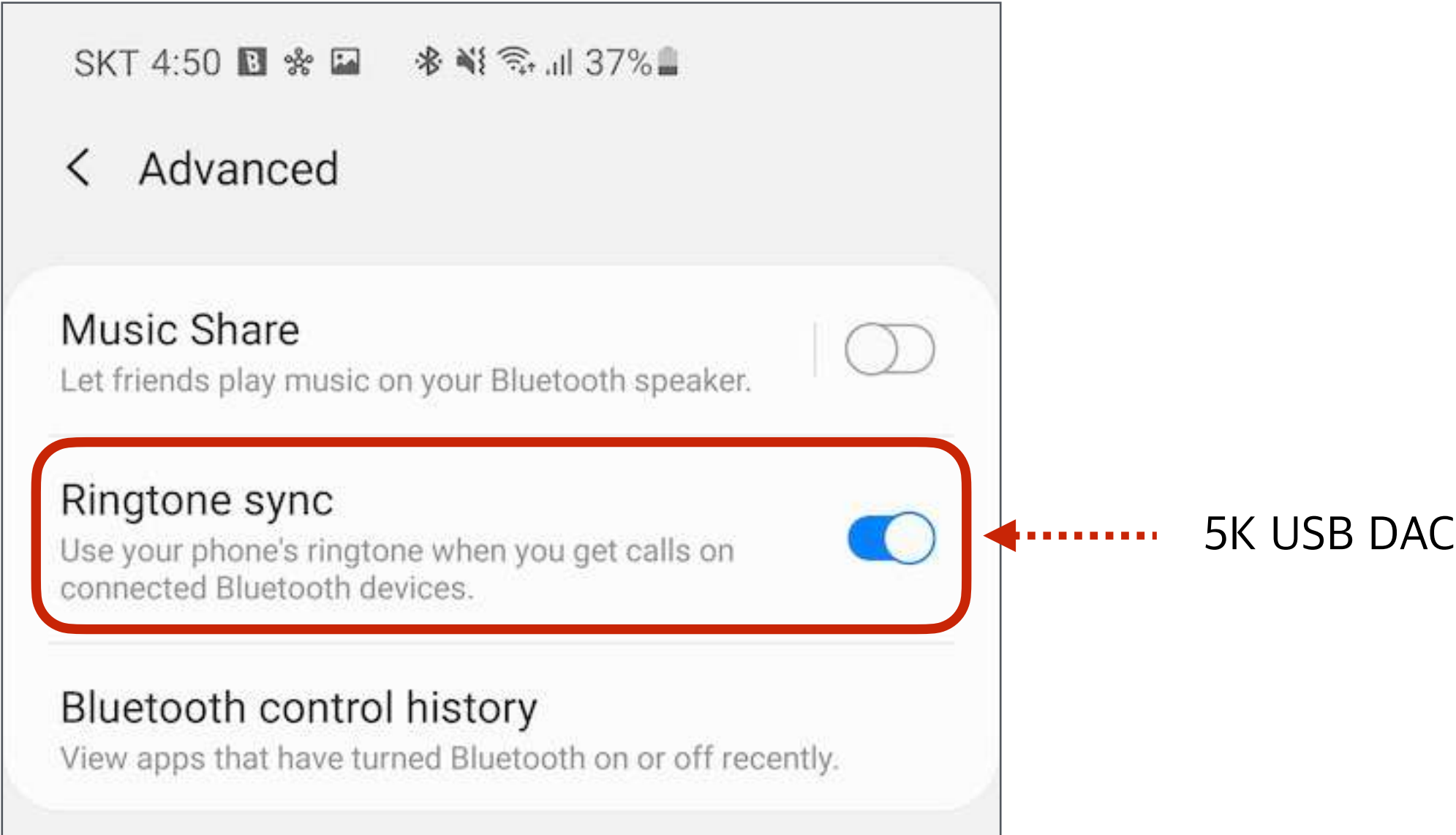
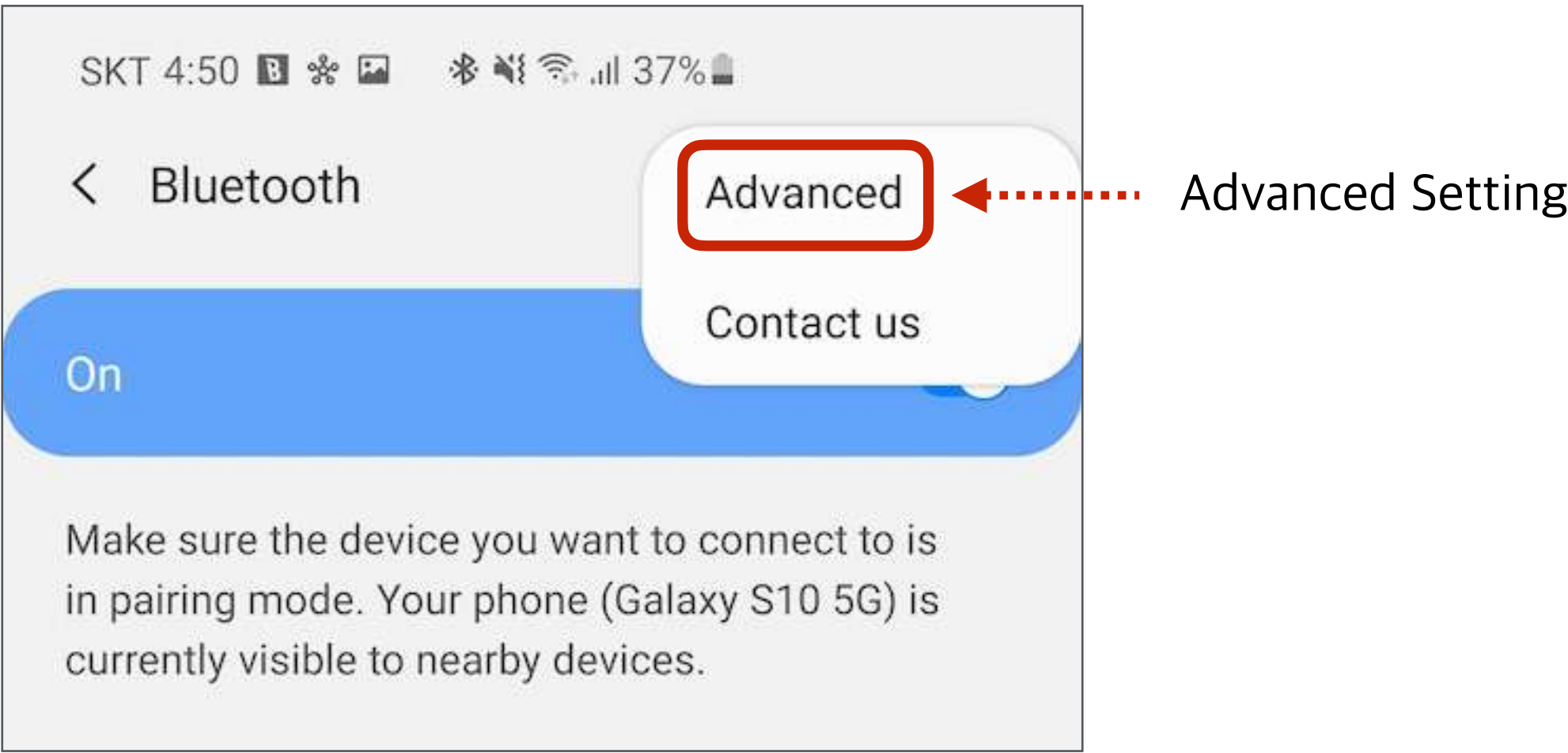
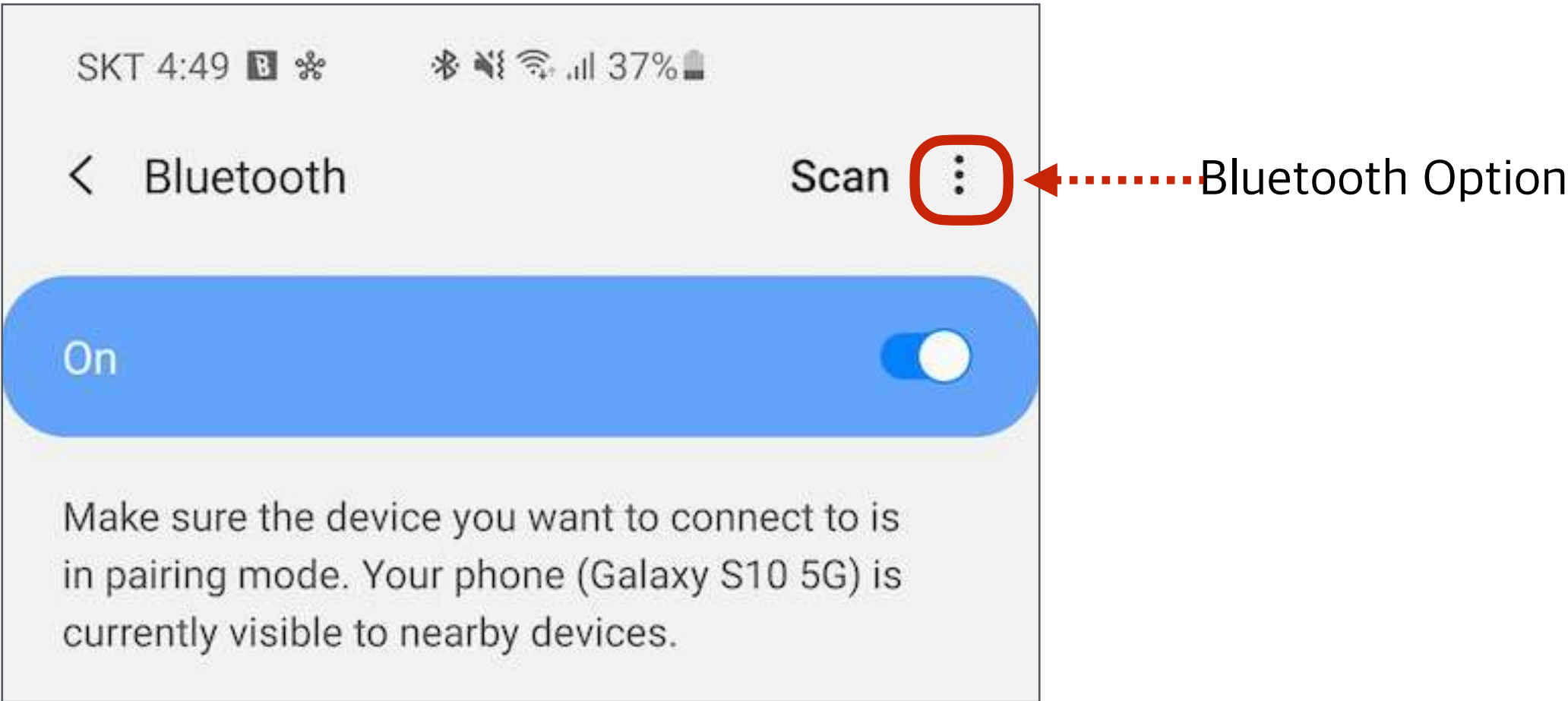
Appendix - iOS Audio Output Device

- 5K can be connected to iPhone both through Bluetooth and USB DAC (Apple Camera Kit) at the same time.
- In that case, the iPhone detects multiple audio output devices. And the iOS selects the active output device exclusively.



Appendix - Android In-Band Ringtone

- The way the Bluetooth receiver plays the incoming call ringtone, is controlled by Android OS.
- System Bluetooth —> Option (Top Right) —> Advanced —> Ringtone sync



Android In-Band Ringtone Galaxy S10